Relativistic Scalar–Vector Plenum: Mathematical Foundations of Lamphrodynamic Field Theory

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Preface

This volume is the first of a three–volume monograph developing the Relativistic Scalar–Vector Plenum (RSVP) as a rigorous mathematical framework. Its primary audience is mathematicians and theoretical physicists with a background in differential geometry, category theory, and quantum field theory. A secondary aim is pedagogical: the material is written so that physicists willing to learn derived geometry can follow, provided they are comfortable with homological algebra and gauge theory.

The central goal of Volume I is to construct the lamphrodynamic field theory underlying RSVP, to formulate it in the language of derived algebraic geometry and shifted symplectic structures, and to establish fundamental analytic results (well–posedness, energy estimates, regularity). Later volumes will build on this foundation to explore cosmology, emergent gravity, geometric computation, and philosophical implications.

We have tried to adhere to the following design principles:

- *Rigor first in foundations.* The core constructions (derived stacks, shifted symplectic forms, AKSZ action) are presented with as much mathematical rigor as possible.
- *Motivation before formalism.* Each chapter begins with an informal explanation of why the subsequent technical machinery is needed.
- *Explicit status of claims.* Theorems are proved or referenced; conjectures and open problems are clearly labeled and accompanied by evidence or heuristic arguments.

Volume I is structured in four parts. Part I reviews the necessary background in derived geometry and shifted symplectic structures, oriented toward physicists. Part II introduces the lamphrodynamic plenum and constructs the relevant derived moduli stacks and AKSZ action. Part III develops the analytic foundations of the lamphrodynamic PDEs. Part IV investigates geometric and topological properties, including entropic singularities and symmetries.

Subsequent volumes will not repeat this material, but will refer back to it frequently.

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Contents

Notation and Conventions

Unless otherwise stated, we adopt the following conventions:

- (M, g) denotes a smooth oriented Lorentzian or Riemannian manifold, with metric g of signature $(-, +, \dots, +)$ in the Lorentzian case.
- Greek indices $\mu, \nu, \dots$ range over spacetime coordinates; Latin indices $i, j, \dots$ are reserved for spatial coordinates when a splitting is chosen. We freely use the abstract index notation of Penrose.
- The Levi–Civita connection of g is denoted by ∇ and the associated Laplace–Beltrami operator by $\Delta = \nabla_a \nabla^a$.
- The symbol $\mathrm{Map}(X, Y)$ denotes a (derived) mapping stack when X and Y are derived stacks, and an ordinary mapping space when they are topological or smooth manifolds; the intended sense is clear from context.
- All tensor contractions use the Einstein summation convention.
- The core lamphrodynamic fields of RSVP are denoted $(\Phi, \boldsymbol{v}, S)$: a scalar potential, a vector flow, and an entropy density. We often group them into a single configuration field $\Psi := (\Phi, \boldsymbol{v}, S)$.

We recommend that readers unfamiliar with derived stacks or shifted symplectic geometry skim Chapters 1 and 2 first, and then consult Chapter 3 and **??** as needed when these notions appear later.

Part I

Derived Algebraic Geometry for Physicists

Chapter 1

Introduction and Motivation

The Relativistic Scalar–Vector Plenum (RSVP) is a proposal for a unified field–theoretic framework in which gravity, cosmology, and certain aspects of information and agency arise from the dynamics of a triplet of fields

$$(\Phi, \boldsymbol{v}, S)$$

living on a spacetime manifold (M, g) or, more generally, on a derived geometric object that refines (M, g). The scalar field Φ encodes a form of lamphrodynamic potential, $\boldsymbol{v}$ represents a flux or flow of negentropic influence, and S is an entropy density that couples to geometry and matter.

At a heuristic level, RSVP treats spacetime not as a static stage on which fields live, but as a *plenum* whose local structure is determined by the coupled evolution of $(\Phi, \boldsymbol{v}, S)$. Geometrical quantities such as curvature, torsion, and metric relations are then derived from the lamphrodynamic configuration rather than posited *a priori*.

This introductory chapter explains why such a framework naturally leads us to derived algebraic geometry and shifted symplectic structures, and sets the stage for the rigorous constructions that follow.

1.1 From Classical Fields to Derived Geometry

In classical field theory, a field configuration is a section of some bundle over spacetime M, and the space of fields is modeled as an infinite dimensional manifold or Fréchet space. Functional integrals, symmetries, and constraints are handled with varying degrees of rigor depending on context.

However, many modern developments — such as moduli spaces of solutions to gauge equations, intersection theory on spaces of instantons, and the BV formalism for gauge theories — reveal that naive smooth manifolds are insufficient. One encounters:

- Singularities when solutions collide or degenerate.
- Higher automorphisms from gauge symmetries.
- Obstructions and derived intersections.

Derived algebraic geometry was developed precisely to give these moduli spaces a well–behaved home. Instead of regarding a moduli space as a naive quotient or set of solutions, one encodes it as a derived stack equipped with additional structure (cotangent complex, obstruction theory, shifted symplectic form).

Intuition. Even if the base spacetime M is an ordinary smooth manifold, the *space of fields* is almost never a manifold in the classical sense. Gauge symmetry, constraints, and singularities force us to work in a homotopical setting. Derived stacks and ∞–categories are the natural language for this.

In RSVP, lamphrodynamic fields are subject to:

- Gauge transformations mixing Φ and $\boldsymbol{v}$,
- Constraints relating S to curvature and matter content,
- Entropic singularities where S becomes non–smooth or diverges.

These features suggest that the moduli of lamphrodynamic configurations should be modeled as a derived stack, which we denote Plenum. A significant portion of Volume I is devoted to constructing Plenum, equipping it with a (-1)–shifted symplectic structure, and understanding its tangent and obstruction theory.

1.2 The Lamphrodynamic Triplet

We give a preliminary, heuristic description of the fields before formalizing them in **????**.

- Φ is a real–valued scalar field on M (or a section of some line bundle) representing a lamphrodynamic potential. Regions of high Φ can be thought of as reservoirs of "available negentropy".
- $\boldsymbol{v}$ is a vector field on M, or more generally a section of $TM \otimes E$ for some auxiliary bundle E, encoding the directed transport of lamphrodynamic influence. It plays a role analogous to a current or velocity field.
- S is a scalar entropy density that couples to both geometry and matter fields. In contrast to thermodynamic entropy defined on macrostates, S is intended as a local field that participates dynamically in the evolution equations.

The basic lamphrodynamic field equations are schematically of the form

$$\partial_t \Phi = \mathcal{D}_\Phi \Delta \Phi + \mathcal{F}_\Phi(\Phi, \boldsymbol{v}, S, g), \tag{1.1}$$

$$\partial_t S = \mathcal{D}_S \Delta S + \mathcal{F}_S(\Phi, \boldsymbol{v}, S, g), \tag{1.2}$$

$$\partial_t \boldsymbol{v} = \mathcal{D}_v \Delta \boldsymbol{v} + \mathcal{F}_v(\Phi, \boldsymbol{v}, S, g), \tag{1.3}$$

where $\mathcal{D}_\Phi, \mathcal{D}_S, \mathcal{D}_v$ are diffusion constants and the nonlinear terms $\mathcal{F}_{\Phi,S,v}$ encode lamphrodynamic interactions and coupling to curvature. Precise versions of these equations will be derived from an action principle in **????**.

Warning. Equations (1.1)–(1.3) are merely schematic at this stage. They suggest that RSVP is, at its analytic core, a system of parabolic or mixed–type PDEs; however, the actual lamphrodynamic equations will include nonlocal and higher–order terms arising from the BV–AKSZ construction.

1.3 Why a (-1)–Shifted Symplectic Structure?

In classical Hamiltonian mechanics, the phase space of a system is a symplectic manifold (X, ω) and the dynamics are encoded by a Hamiltonian function H via the equation

$$\iota_{X_H}\omega = \mathrm{d}H.$$

For gauge theories and field theories, the BV formalism generalizes this picture to a graded setting: one replaces (X, ω) by a graded manifold equipped with an odd symplectic form, and the Hamiltonian H by a master action S satisfying $\{S, S\} = 0$.

In derived geometry, these graded symplectic manifolds are captured by *shifted symplectic structures* on derived stacks. A k–shifted symplectic structure is, roughly, a closed, nondegenerate 2–form of cohomological degree k on the derived stack. The case $k = -1$ is particularly important for the BV formalism: it corresponds to odd symplectic forms and is tailored to encode gauge theories.

Definition 1.1. Let $\mathcal{X}$ be a derived Artin stack over a field of characteristic zero. A (-1)*–shifted symplectic structure* on $\mathcal{X}$ is a closed 2–form

$$\omega \in \mathcal{A}^{2,cl}(\mathcal{X})[-1]$$

whose underlying pairing on the cotangent complex is nondegenerate in the derived sense.

The key insight for RSVP is that the moduli stack Plenum of lamphrodynamic configurations should carry such a (-1)–shifted symplectic structure, with the master action functional S_{RSVP} playing the role of a Hamiltonian satisfying the classical master equation

$$\{S_{RSVP}, S_{RSVP}\} = 0.$$

Once this structure is constructed, the AKSZ formalism provides a natural way to build S_{RSVP} from geometric data associated to Plenum and the spacetime source manifold.

Intuition. The choice of $k = -1$ is not arbitrary: lamphrodynamic fields are subject to gauge redundancies and constraints, so their moduli naturally form a BV phase space rather than a naive symplectic manifold. This forces the symplectic structure to be of odd degree, hence $k = -1$.

A central technical goal of Volume I is to:

- Construct Plenum as a derived Artin stack.
- Equip Plenum with a (-1)–shifted symplectic structure ω.

- Show that ω can be expressed in terms of $(\Phi, \boldsymbol{v}, S)$ and curvature data, and that it is closed and nondegenerate.

These results will be established in **????**.

1.4 Analytic Questions: WellPosedness and Regularity

The derived geometric formulation of RSVP provides a powerful conceptual framework, but physics also demands that the associated PDEs be analytically well–behaved. Concretely, we must address:

1. *Well–posedness.* Given initial data $(\Phi_0, \boldsymbol{v}_0, S_0)$, does there exist a unique solution $(\Phi, \boldsymbol{v}, S)$, at least for short time, in appropriate Sobolev spaces?
2. *Energy estimates.* Are there a priori bounds controlling the growth of norms of the fields, and do these bounds depend continuously on the initial data?
3. *Regularity and singularities.* Under what conditions can singularities form in finite time, and how are such singularities related to entropic phenomena in RSVP?
4. *Long–time behavior.* Do solutions converge to lamphrodynamic equilibria or attractors, and can one classify these asymptotic states?

These questions are addressed in Part III of this volume. We will prove global well–posedness for linearized lamphrodynamic equations and establish energy estimates and local well–posedness for certain nonlinear regimes. Global existence for the fully nonlinear system remains conjectural; we will state precise open problems and present partial evidence.

1.5 Organization of the Three Volumes

For orientation, we briefly summarize the roles of the three volumes.

Volume I: Mathematical Foundations

This volume develops:

- The background in derived algebraic geometry and shifted symplectic structures needed to formulate RSVP rigorously.
- The construction of the lamphrodynamic moduli stack Plenum and its (-1)–shifted symplectic structure.
- The AKSZ–BV action for lamphrodynamic fields and its classical master equation.
- Fundamental analytic results on well–posedness, energy estimates, and regularity of the lamphrodynamic PDEs.

Volume II: Physical Applications

The second volume applies this foundation to physics:

- Derivation of modified Einstein equations with an explicit lamphrodynamic stress–energy contribution Θ_{ab}.
- Development of a cosmological model in which redshift, structure formation, and cosmic microwave background anisotropies are governed by lamphrodynamic fields rather than pure metric expansion.
- Detailed comparison with observational data (supernovae, BAO, Planck CMB measurements) and identification of potential tests that distinguish RSVP from ΛCDM.

Volume III: Extensions and Interpretations

The third volume explores:

- The SpherePop calculus as a geometric model of computation, including categorical and homotopical formalization and a proof of Turing completeness.
- L–systems, semantic fields, and distributed computation (CRDTs) as instances of lamphrodynamic or related field theories.
- Entropy–weighted spectral geometry and "cymatic" resonance modes.
- Philosophical and creative dimensions, including the interpretation of RSVP within structural realism and process philosophy.

1.6 What is Proven, What is Conjectural

Given the ambitious scope of RSVP, it is important to distinguish carefully between results that are established in this monograph and those that remain conjectural.

- *Proven in Volume I:* We provide complete proofs for the construction of Plenum as a derived Artin stack, for the existence of a (-1)–shifted symplectic form on Plenum under specified conditions, and for well–posedness results in the linear and certain nonlinear lamphrodynamic regimes.
- *Proven in Volume II:* We derive modified Einstein equations from the lamphrodynamic action, obtain concrete formulas for cosmological observables (including a redshift function $f(S, \boldsymbol{v})$), and perform quantitative comparisons with existing data. The degree of observational agreement is investigated but not treated as a fully conclusive fit.
- *Proven in Volume III:* We rigorously formalize SpherePop as a homotopy–theoretic model of computation, prove that merge operations are homotopy colimits, and establish Turing completeness via explicit encodings.

- *Conjectural:* Global existence of solutions for the full nonlinear lamphrodynamic system; completeness of the cosmological model as a full replacement for ΛCDM; certain speculative connections between lamphrodynamic spectral modes and cognition or consciousness.

These conjectural aspects are clearly labeled as such, and in each case we provide reasons for their plausibility and suggest directions for future research.

1.7 How to Read this Volume

Different readers may wish to approach the material in different orders.

- **Mathematical physicists** familiar with homological algebra and derived geometry may skim Part I and focus on Parts II and III, where the lamphrodynamic stack and PDE analysis are developed.
- **Physicists new to derived geometry** are encouraged to read Chapters 2 and 3 carefully, referring to appendices as needed, before tackling the AKSZ construction in **????**.
- **Analysts** interested in PDE aspects may wish to jump directly to Part III, using the early chapters primarily for motivation and notation.

Throughout, we have tried to separate background material, core constructions, and speculative interpretations. The reader is invited to treat this volume as a foundation that can be built upon independently of whether they ultimately accept the full physical picture proposed by RSVP.

In the next chapter, we begin our technical journey by reviewing the basic language of model categories, ∞–categories, and derived stacks, with an eye toward their eventual application to lamphrodynamic field theory.

Chapter 2

∞–Categories and Homotopy Theory

This chapter reviews the homotopical and ∞–categorical structures needed to formulate derived geometry and, ultimately, the lamphrodynamic moduli stack Plenum developed in later chapters. We assume that the reader is familiar with basic category theory and classical homotopy theory. Our goal is not to give a comprehensive treatment, but to present a minimal collection of definitions, constructions, and results that will be invoked in the sequel.

Throughout, we work over a field of characteristic zero (typically $\mathbb{R}$ or $\mathbb{C}$) unless otherwise specified. References include [**Lurie2009HTT**], [**Lurie2017HA**], and [**ToenVezzosi2008**].

2.1 Model Categories and Homotopical Structures

The motivation for ∞–categories begins with the desire to regard homotopy equivalent objects as "the same" while retaining control over higher homotopies. Classical category theory does not remember higher homotopies; instead, it collapses homotopies to identities. In homotopy theory, however, we need to keep track of:

- homotopies between maps,
- homotopies between homotopies,
- and so on, ad infinitum.

A traditional approach is provided by *model categories*, which endow a category with distinguished classes of weak equivalences, fibrations, and cofibrations satisfying Quillen's axioms [**Quillen1967**]. The basic example is the model structure on topological spaces, where weak equivalences are weak homotopy equivalences, fibrations are Serre fibrations, and cofibrations are retracts of relative CW–complex inclusions.

While model categories provide a powerful toolkit, they are ultimately an *encoding* of ∞–categorical structures, and the language of ∞–categories gives a more conceptual, coordinate–free approach.

2.2 ∞–Categories

An ∞–category is a category enriched in spaces rather than sets, or, equivalently, a homotopical object that remembers all higher morphisms. A concrete model is given by *quasi–categories* (simplicial sets satisfying inner horn conditions) introduced by Boardman and Vogt and developed extensively by Joyal and Lurie.

Definition 2.1. A *quasi–category* is a simplicial set C such that every inner horn $\Lambda^n_i \to C$ admits a filler $\Delta^n \to C$ for $0 < i < n$.

This condition generalizes the Kan condition for simplicial sets corresponding to ∞–groupoids (spaces). Every topological space has a singular complex which is a Kan complex; every Kan complex is a model for an ∞–groupoid. The quasi–category condition weakens Kan filling so that objects need not have all inverses, thus modeling ∞–categories rather than ∞–groupoids.

Intuition. Ordinary categories forget higher homotopies between morphisms. Quasi–categories remember not only morphisms but also homotopies and homotopies between homotopies, all the way up. This makes them ideal for encoding moduli spaces, where higher automorphisms naturally appear.

2.3 ∞–Topoi and Sheaves of Spaces

Derived geometry requires a setting where "sheaves of homotopical data" can be treated systematically. This motivates the notion of an ∞–topos: a homotopical enhancement of Grothendieck topoi.

Definition 2.2. An *∞–topos* is an ∞–category that is equivalent to an ∞–category of sheaves of spaces on a Grothendieck site. Equivalently, it is a presentable ∞–category satisfying descent for hypercovers [**Lurie2009HTT**].

Classical sheaf theory associates sets or vector spaces to open subsets of a manifold. An ∞–topos associates *spaces* (i.e. ∞–groupoids) to open subsets, thereby encoding higher homotopical information. Derived stacks are then defined as (higher) sheaves of ∞–groupoids satisfying appropriate representability and local geometricity conditions.

Remark 2.3. The ∞–topos of sheaves of spaces on the site $(\mathrm{Sch}, \mathrm{Zariski})$ is denoted $\mathrm{Shv}_\infty(\mathrm{Sch})$. Derived algebraic geometry works mostly inside suitable subcategories of this ∞–topos.

2.4 Derived Stacks

Derived stacks were developed by ToënVezzosi, Lurie, and others to handle moduli problems with obstructions, singularities, and nontrivial automorphism groups. Roughly speaking, derived stacks are functors from derived rings to ∞–groupoids satisfying descent and representability conditions.

Definition 2.4. A *derived stack* is a functor

$$F : \mathrm{dAlg}^{op} \longrightarrow \infty\text{–Gpd}$$

satisfying ∞–geometricity and descent conditions.

Derived stacks generalize classical schemes and algebraic stacks, adding:

- derived intersections,
- cotangent complexes,
- obstruction theories,
- and higher automorphisms.

Intuition. In physics, moduli spaces of solutions to gauge equations (e.g. Yang–Mills, GR) often have singularities and higher gauge symmetries. Derived stacks treat these automatically, removing the need for ad hoc patching arguments.

In RSVP, the moduli stack Plenum of lamphrodynamic configurations will be constructed as a derived Artin stack equipped with a (-1)–shifted symplectic structure. This stack encodes field configurations, gauge orbits, and entropic singularities in a unified homotopical object.

2.5 Tangent Complexes and Obstruction Theory

Classical manifolds possess tangent spaces at each point, controlling infinitesimal deformations. Derived stacks generalize this to a *tangent complex*, which is typically a chain complex, possibly with cohomology in multiple degrees.

Definition 2.5. Let $x \in F$ be a point of a derived stack F. The *tangent complex* T_xF is the derived mapping space

$$T_xF := \mathrm{Map}_{\mathrm{dStk}}(\mathrm{Spec}\, k[\epsilon]/(\epsilon^2), F)_x,$$

considered as a complex in degrees $(-\infty, \ldots, \infty)$.

The cohomology groups of T_xF encode:

- $H^0(T_xF)$ classical tangent directions,
- $H^{-1}(T_xF)$ infinitesimal automorphisms,
- $H^1(T_xF)$ obstructions to extending deformations,
- higher degrees controlling higher obstructions and symmetries.

In particular, if F is smooth (in derived sense), then $H^1(T_xF)$ may vanish, ensuring unobstructed deformations; if $H^{-1}(T_xF)$ is nonzero, x has nontrivial automorphisms (gauge symmetries).

Intuition. The tangent complex of Plenum will encode:

- infinitesimal variations of $(\Phi, \boldsymbol{v}, S)$,
- gauge symmetries mixing Φ and $\boldsymbol{v}$,
- obstructions related to entropic singularities.

2.6 Cotangent Complex and Shifted Symplectic Structures

The cotangent complex L_F of a derived stack F is a fundamental invariant generalizing the classical cotangent bundle. It plays a central role in the definition of shifted symplectic structures.

Definition 2.6. The *cotangent complex* L_F is a perfect complex on F (when F is derived Artin) whose dual $T_F := L_F^\vee$ generalizes the tangent bundle.

Shifted symplectic structures are defined as certain closed bilinear forms on L_F (or T_F) of cohomological degree k. The case $k = -1$ is the setting of the BV formalism and will be central to RSVP.

We discuss shifted symplectic structures in detail in Chapter 3.

2.7 ∞–Groupoids and Fields

In ordinary geometry, a point in a moduli space of fields corresponds to a single classical solution. In derived geometry, such points carry higher homotopical structure: their automorphism groups form ∞–groupoids, which encode gauge equivalences and higher symmetries.

Consequently, the moduli stack Plenum should be thought of as an object in an ∞–category of derived stacks, with points representing field configurations, morphisms representing gauge transformations, and higher morphisms encoding redundancy among transformations themselves.

This viewpoint is essential to the BV formalism: ghosts, antifields, and the master equation all arise naturally from the derived structure of Plenum.

2.8 Summary and Outlook

This chapter introduced the basic machinery underlying derived geometry: model categories, ∞–categories, ∞–topoi, and derived stacks, with an emphasis on tangent and cotangent complexes and higher automorphisms. These tools will be used in Chapter 3 and **??** to define and analyze shifted symplectic structures and the AKSZ formalism.

In **??**, we will construct the lamphrodynamic moduli stack Plenum explicitly as a derived Artin stack, then equip it with a (-1)–shifted symplectic form in **??**. The present chapter thus serves as the homotopical foundation for all subsequent developments.

Chapter 3

Shifted Symplectic Structures

This chapter develops the basic notions of shifted symplectic geometry used to formulate the moduli stack Plenum of lamphrodynamic configurations as a (-1)–shifted symplectic derived Artin stack. We assume familiarity with the cotangent complex of a derived stack and its dual tangent complex as introduced in Chapter 2. Throughout, we follow [**PTVV2013**], [**Calaque2015**] and related sources.

3.1 Preliminaries: Classical Symplectic Geometry

We begin with a brief review of classical symplectic geometry to fix notation and highlight the generalization to derived settings.

Definition 3.1. A *symplectic manifold* is a pair (X, ω) where X is a smooth manifold and $\omega \in \Omega^2(X)$ is a closed (i.e. $\mathrm{d}\omega = 0$) and nondegenerate 2–form.

A central example is T^*Q for a smooth manifold Q, equipped with the canonical 2–form $\omega = \mathrm{d}p_i \wedge \mathrm{d}q^i$. Classical Hamiltonian mechanics is framed on (X, ω) with Hamiltonian $H \in C^\infty(X)$.

In gauge theory and in field theory, however, the moduli space of classical solutions is typically singular, infinite dimensional, and equipped with higher automorphisms, so classical manifolds are no longer adequate. A homotopical refinement is needed, leading to the notion of shifted symplectic structures.

3.2 Graded Geometry and Cohomological Degree

In derived geometry, differential forms are replaced by cohomological complexes and their degrees must be tracked carefully. A k–shifted symplectic form is, informally, a closed 2–form of cohomological degree k on the derived stack. Heuristically:

$$\omega \in H^k(\mathcal{A}^{2,cl}(F)),$$

where F is a derived Artin stack and $\mathcal{A}^{2,cl}(F)$ is the sheaf of closed 2–forms on F.

Definition 3.2. Let F be a derived Artin stack. A *k–shifted symplectic structure* on F is a closed 2–form ω of cohomological degree k, represented by a morphism of complexes

$$\omega : \wedge^2 L_F \longrightarrow \mathcal{O}_F[k],$$

where L_F is the cotangent complex of F, such that the induced pairing $L_F^\vee \otimes L_F^\vee \to \mathcal{O}_F[k]$ is nondegenerate in the derived sense.

Here $L_F^\vee = T_F$ generalizes the tangent bundle. The condition that L_F be perfect (locally of finite Tor dimension) ensures that T_F behaves well as a dual.

Intuition. The shift k controls the "oddness" or "evenness" of the symplectic form. Classical symplectic structures correspond to $k = 0$. The case $k = -1$ corresponds to odd symplectic geometry and plays the role of BV phase spaces.

3.3 (-1)–Shifted Symplectic Structures and BV

The case $k = -1$ is the setting of the BatalinVilkovisky (BV) formalism for gauge theories. In this context, the derived stack F encodes classical fields and gauge symmetries; the shifted symplectic form ω is odd; and the BV bracket

$$\{-,-\} : \mathcal{O}_F \otimes \mathcal{O}_F \to \mathcal{O}_F[-(-1)] = \mathcal{O}_F[1]$$

is defined from ω.

Definition 3.3. Given a (-1)–shifted symplectic structure ω on F, the *BatalinVilkovisky bracket* $\{-,-\}$ on functions (or, more precisely, on appropriate cochains) is the unique odd Poisson bracket induced by ω.

The BV mechanism encodes gauge symmetries as degenerate directions of the action, formally resolved by ghosts and higher homotopies. The *classical master equation*

$$\{S, S\} = 0$$

then expresses invariance under these symmetries. Later, in **????**, we will show how the lamphrodynamic action S_{RSVP} arises from an AKSZ construction on the moduli stack Plenum equipped with a (-1)–shifted symplectic structure.

3.4 Closedness and Nondegeneracy in the Derived Sense

Closedness of ω is expressed in terms of the differential and the cohomological structure on L_F. Nondegeneracy means that the induced morphism

$$T_F \longrightarrow L_F[-k]$$

is an isomorphism in the derived category. For $k = -1$, this becomes

$$T_F \longrightarrow L_F[1],$$

which is automatically compatible with the BV grade.

Remark 3.4. For $k < 0$, the shifted symplectic form is “odd” and induces higher order compatibilities among tangent and cotangent complexes, matching precisely the structure required for the BV bracket.

Closedness means that ω lifts to a cocycle in a suitable truncated de Rham complex on F. We do not reproduce the full homotopical definition here (see [**PTVV2013**]), but emphasize that closedness is a global condition involving sheaf descent and higher coherences.

3.5 Examples and Fundamental Results

Key results from PantevToënVaquiéVezzosi [**PTVV2013**] include:

Theorem 3.5 (PTVV)**.** *Let X be a smooth scheme and let T^*kX be the k–shifted cotangent stack. Then T^*kX carries a canonical k–shifted symplectic structure.*

This theorem provides the fundamental example of shifted symplectic structures and plays a role analogous to the cotangent bundle in classical mechanics. In derived geometry, one generalizes the phase space to T^*kX for any integer k, with $k = -1$ corresponding to BV theory.

Another key result concerns mapping stacks:

Theorem 3.6 (PTVV)**.** *Let M be an oriented smooth manifold of dimension d, and let F be a derived Artin stack with an n–shifted symplectic structure. Then the mapping stack* $\mathrm{Map}(M, F)$ *carries an $(n - d)$–shifted symplectic structure.*

This result is central to the AKSZ construction, where M is typically the source manifold (e.g. the spacetime worldvolume) and F is the target stack encoding the fields. For RSVP, M will often be spacetime or a spacelike hypersurface, and F the lamphrodynamic stack Plenum.

Intuition. Shifted symplecticity propagates from target to mapping stack, reducing the shift by the dimension of the domain. For a 3–dimensional worldvolume and an n–shifted target, the resulting structure is $(n - 3)$–shifted.

3.6 Shifted Symplectic Forms from Field Content

In RSVP, the (-1)–shifted symplectic structure arises from the lamphrodynamic triplet $(\Phi, \boldsymbol{v}, S)$ and their coupling to geometry. Roughly speaking, the moduli stack Plenum of field configurations admits a cotangent complex generated by variations of the fields. Pairing these variations through curvature and entropy flow produces a candidate closed 3–form, whose degree and symmetries align with $k = -1$.

In **??**, we will:

1. Construct Plenum as a derived Artin stack.

2. Identify its cotangent complex L_{Plenum} explicitly in terms of variations of $(\Phi, \boldsymbol{v}, S)$.

3. Construct a (-1)–shifted symplectic form ω_{Plenum} expressed locally as a derived 3–form built from the lamphrodynamic fields.

4. Verify that ω_{Plenum} is closed and nondegenerate.

This is a central technical achievement of Volume I.

3.7 Why $k = -1$?

A natural question is: why specifically $k = -1$ for RSVP? Why not $k = 0$ or $k = -2$?

- The case $k = 0$ would produce an *even* symplectic form. This is appropriate for classical mechanics or certain topological sigma models, but not for gauge theories with ghost content.

- The BV formalism requires an odd symplectic form so that the BV bracket has the correct parity. This forces k to be odd and negative.

Moreover, gauge symmetries and redundancy among lamphrodynamic fields indicate that Plenum is not a naive moduli space but a derived stack with nontrivial homotopy in negative degrees. The canonical choice compatible with BV is $k = -1$, aligning with the degree of the ghost sector and with the reductions arising in AKSZ.

Remark 3.7. In some exotic generalizations, higher negative shifts may appear (e.g. topological field theories or higher spin models). For RSVP, the entropic and lamphrodynamic structure strongly suggest $k = -1$ as the natural choice.

3.8 Shifted Symplecticity and AKSZ

The AKSZ construction, introduced in [**AKSZ**], produces solutions of the classical master equation from a target derived stack equipped with a shifted symplectic structure. The recipe is:

1. Choose a source manifold M (e.g. spacetime).

2. Choose a target derived stack F with an n–shifted symplectic structure.

3. Form the mapping stack $\mathrm{Map}(M, F)$; by PTVV, this inherits an $(n - \dim M)$–shifted symplectic structure.

4. For $n - \dim M = -1$, obtain an odd symplectic structure on $\mathrm{Map}(M, F)$, suitable for BV quantization.

In RSVP, F will ultimately be the lamphrodynamic target Plenum and M the spacetime manifold. The resulting (-1)–shifted symplectic structure on $\mathrm{Map}(M, \mathsf{Plenum})$ provides the geometric foundation for the BV action S_{RSVP}.

3.9 Summary and Forward Outlook

Shifted symplectic geometry generalizes classical symplectic structures to a homotopical context and provides the correct language for BV and AKSZ quantization. The case $k = -1$ corresponds precisely to odd symplectic geometry required for lamphrodynamic gauge theory.

In the next chapter, **??**, we will review the AKSZ formalism in detail, emphasizing how shifted symplectic structures on target stacks determine BV actions, and how the constraint $n - \dim M = -1$ leads naturally to lamphrodynamic dynamics on Plenum.

Chapter 4

AKSZ Formalism and the Classical BV Construction

4.1 Introduction and Motivation

This chapter introduces the Alexandrov–Kontsevich–Schwarz–Zaboronsky (AKSZ) construction for classical field theories with a $(k-1)$-shifted symplectic target, following [**CF01**; **CF07**; Ale+97; Pan+13]. The formalism generalizes the classical BatalinVilkovisky (BV) method and provides a canonical route to constructing solutions of the classical master equation. The approach is intrinsically geometric, relying on mapping stacks, graded manifolds, and homological vector fields of degree $+1$.

From the perspective of RSVP, the AKSZ framework is essential. RSVP fields will be realized as maps

$$\mathcal{F} \colon \mathsf{T}[1]M \longrightarrow \mathcal{X},$$

where $\mathcal{X}$ is the derived moduli stack of lamphrodynamic triples $(\Phi, \vec{v}, S)$ constructed in Chapter 6, equipped with a canonical (-1)-shifted symplectic structure. The AKSZ recipe then produces a BV action functional S_{BV} satisfying $\{S_{\mathrm{BV}}, S_{\mathrm{BV}}\} = 0$, thereby yielding a consistent classical theory in the BV sense.

This chapter develops the required formal background, establishes the general AKSZ construction, and explains the classical BV complex. The final section illustrates the formalism with the example of three-dimensional Chern–Simons theory.

Pedagogical note. The first half of this chapter is written from the perspective of a physicist learning derived geometry; the second half assumes familiarity with shifted symplectic structures, homological vector fields, and the BV spectral sequence.

4.2 Graded Manifolds and Homological Vector Fields

Let $\mathcal{M}$ be a $\mathbb{Z}$-graded supermanifold (or more generally, a derived stack locally modeled by such). A *homological vector field* is a degree $+1$ vector field Q on $\mathcal{M}$ satisfying

$$Q^2 = \tfrac{1}{2}[Q, Q] = 0.$$

The pair $(\mathcal{M}, Q)$ is called a *Q-manifold*. The condition $Q^2 = 0$ implies that Q determines a differential on the algebra of functions $\mathcal{O}_{\mathcal{M}}$, and hence gives rise to a cochain complex $(\mathcal{O}_{\mathcal{M}}, Q)$.

Definition 4.1. A *Q-mapping stack* from $\mathsf{T}[1]M$ to $\mathcal{X}$ is a mapping stack

$$\mathrm{Map}(\mathsf{T}[1]M, \mathcal{X})$$

equipped with a homological vector field induced from the vector fields on both source and target, in particular the de Rham differential on $\mathsf{T}[1]M$.

Remark 4.2. Physically, the degree shift encodes ghost number. Coordinates of degree 0 describe classical fields, degree -1 antifields, degree $+1$ ghosts.

4.3 Shifted Symplectic Structures and the BV Bracket

Let $(\mathcal{X}, \omega)$ be a k-shifted symplectic derived stack. Thus ω is a closed, non-degenerate 2-form of cohomological degree k. When $k = -1$, the induced Poisson bracket has cohomological degree $+1$, and is known as the BV bracket.

Definition 4.3. Given a (-1)-shifted symplectic structure ω, the *BV bracket* $\{-,-\}$ on $\mathcal{O}_{\mathcal{X}}$ is defined by

$$\{f, g\} := \iota_{X_f} \iota_{X_g} \omega,$$

where X_f is the Hamiltonian vector field determined by f.

Proposition 4.4. *The BV bracket has the following properties:*

1. $\{f, g\} = -(-1)^{(|f|+1)(|g|+1)}\{g, f\}$,
2. $\{f, gh\} = \{f, g\}h + (-1)^{(|f|+1)|g|} g\{f, h\}$,
3. $(-1)^{(|f|+1)(|h|+1)}\{f, \{g, h\}\} + \textit{cyclic perms} = 0$.

Proof (sketch). The identities follow from the graded symplectic structure and the Cartan formula. A complete proof may be found in [**Schwarz93**; **CattaneoFelder02**; Pan+13]. □

4.4 AKSZ Data and the Classical Master Equation

Let M be a smooth oriented manifold of dimension d, and let $(\mathcal{X}, \omega)$ be a $(d-1)$-shifted symplectic target with a homological vector field Q preserving ω. The AKSZ construction associates to this data a BV action on the mapping stack

$$\mathcal{F} := \mathrm{Map}(\mathsf{T}[1]M, \mathcal{X}),$$

with an induced (-1)-shifted symplectic structure. Concretely, the AKSZ action is given by

$$S_{\mathrm{AKSZ}} = \int_{\mathsf{T}[1]M} \left(\langle \Phi^*\theta, d_M \Phi \rangle + \Phi^*\alpha\right), \tag{4.1}$$

where θ is a potential of ω and α is a Hamiltonian function satisfying $Q = \{\alpha, -\}$.

Theorem 4.5 (AKSZ classical master equation)**.** *The functional S_{AKSZ} satisfies*

$$\{S_{\text{AKSZ}}, S_{\text{AKSZ}}\} = 0.$$

Proof (sketch). The construction of S_{AKSZ} is arranged so that the BV bracket encodes the graded symplectic form on the mapping space, and the homological vector field encodes the differential on $\mathsf{T}[1]M$. The fact that Q preserves ω and squares to zero implies that S_{AKSZ} is a solution of the classical master equation. For complete details see [**CF01**; **CF07**; Ale+97; Pan+13]. □

4.5 Mapping Stacks and the Induced Shifted Symplectic Form

Let $(\mathcal{X}, \omega)$ be a $(d-1)$-shifted symplectic derived stack in the sense of [Pan+13], and consider a compact oriented d-manifold M. The AKSZ construction uses the internal hom $\mathrm{Map}(\mathsf{T}[1]M, \mathcal{X})$, viewed as a derived mapping stack in the ∞-topos of derived stacks. The tangent complex is obtained by the standard formula

$$T_\Phi \mathrm{Map}(\mathsf{T}[1]M, \mathcal{X}) \simeq \mathbf{R}\Gamma(\mathsf{T}[1]M, \Phi^* T_{\mathcal{X}}),$$

which identifies tangent vectors at a field configuration Φ with sections of the pullback of the tangent complex of $\mathcal{X}$.

Theorem 4.6 (PTVV)**.** *Let $(\mathcal{X}, \omega)$ be $(d-1)$-shifted symplectic and let M be an oriented d-manifold. Then the mapping stack $\mathrm{Map}(\mathsf{T}[1]M, \mathcal{X})$ carries a canonical (-1)-shifted symplectic structure ω_{AKSZ} induced by integration along M.*

Proof (sketch). The degree shift arises from the Thom form of M and the graded geometry of $\mathsf{T}[1]M$. Integration reduces degree by d, and combined with the $(d-1)$-shift of ω, yields -1. The construction and closedness are established in [Pan+13]. □

Remark 4.7*.* This result is fundamental: any $(d-1)$-shifted symplectic target yields a (-1)-shifted symplectic BV theory on the mapping stack. In particular, the RSVP target constructed in Chapter 6 will have $d = 3$, hence a (-1)-shifted BV theory in four spacetime dimensions.

4.6 BV Fields, Ghosts, and Antifields

Given the AKSZ mapping stack $\mathcal{F} = \mathrm{Map}(\mathsf{T}[1]M, \mathcal{X})$, the derived geometry automatically organizes field components by ghost number. More precisely, a degree-k coordinate on $\mathcal{X}$ pulls back to a degree-$(k+i)$ coordinate on $\mathcal{F}$, where i labels the internal grading of $\mathsf{T}[1]M$. Classical fields occur at total degree 0, ghosts at $+1$, antifields at -1, and higher ghosts at degrees $> +1$.

Definition 4.8. The *BV complex* associated to $\mathcal{F}$ is the cochain complex $(\mathcal{O}_{\mathcal{F}}, Q_{\text{AKSZ}})$, where Q_{AKSZ} is the homological vector field induced from both the target differential and the de Rham differential on the source.

Remark 4.9. In classical physics language, we may write fields schematically as

$$\{\text{fields } \phi,\ \text{ghosts } c,\ \text{antifields } \phi^*,\ \text{anti-ghosts } c^*, \dots\},$$

with degrees organized by the AKSZ grading.

4.7 The Classical BV Action and the Master Equation

Let $(\mathcal{X}, \omega)$ be $(d-1)$-shifted symplectic, and let θ be a potential of ω (i.e. $\omega = d\theta$). Assume $\mathcal{X}$ carries a Hamiltonian function α of degree d such that the associated Hamiltonian vector field $Q = \{\alpha, -\}$ coincides with the homological vector field on $\mathcal{X}$. Then the AKSZ action (4.1) satisfies the BV classical master equation

$$\{S_{\text{AKSZ}}, S_{\text{AKSZ}}\} = 0.$$

This is the precise sense in which AKSZ produces a BV theory.

Proposition 4.10. *The BV bracket associated to ω_{AKSZ} coincides with the variational Schouten bracket on functionals over the mapping stack.*

Proof (sketch). This is essentially the compatibility of the PTVV bracket with the internal Hom structure and the universal property of the mapping stack. See [**CF07**; Pan+13]. □

4.8 Example: Three-Dimensional Chern–Simons Theory

We illustrate the general AKSZ recipe by recovering three-dimensional Chern–Simons gauge theory, following [**CF01**; **CF07**; Ale+97]. Let M be a 3-dimensional oriented manifold and let $\mathfrak{g}$ be a finite-dimensional Lie algebra with invariant bilinear form $\langle \cdot, \cdot \rangle$. Consider the graded vector space

$$\mathfrak{g}[1],$$

concentrated in degree 1, i.e. coordinates have ghost number 1. Equip $\mathfrak{g}[1]$ with the homological vector field

$$Q(x) = \tfrac{1}{2}[x, x],$$

which is of degree $+1$ and squares to zero by the Jacobi identity.

Proposition 4.11. *The space $\mathfrak{g}[1]$ carries a canonical 2-shifted symplectic form (hence $(d-1) = 2$), given by the bilinear form of degree 2 induced from $\langle \cdot, \cdot \rangle$.*

Proof (sketch). The invariant bilinear form provides a non-degenerate pairing of appropriate degree, and closedness follows from the Lie algebra structure. See [Ale+97]. □

Applying the general AKSZ recipe with $d = 3$ and target $\mathfrak{g}[1]$, the mapping stack $\mathrm{Map}(\mathsf{T}[1]M, \mathfrak{g}[1])$ obtains a (-1)-shifted symplectic structure. Let A denote the superfield obtained from the pullback of the graded coordinate on $\mathfrak{g}[1]$; its degree-0 component is an ordinary connection 1-form on M, and higher-degree components encode ghosts and antifields.

The AKSZ action takes the familiar form

$$S_{\mathrm{CS}} = \int_M \left\langle A \wedge dA + \tfrac{1}{3} A \wedge [A, A] \right\rangle, \tag{4.2}$$

where products and brackets are understood in the graded sense. One verifies that $\{S_{\mathrm{CS}}, S_{\mathrm{CS}}\} = 0$ using the graded Jacobi identity and invariance of $\langle \cdot, \cdot \rangle$.

Remark 4.12. The usual Chern–Simons action is the restriction of S_{CS} to the classical degree-zero part of the AKSZ field. All ghosts, antifields, and higher-degree components arise automatically from the AKSZ mapping stack, rather than being introduced by hand.

Forward link to RSVP. In Chapter 7 we apply the AKSZ recipe to the 3-shifted symplectic target Plenum of the lamphrodynamic triples $(\Phi, \vec{v}, S)$, constructed explicitly in Chapter 6. Because Plenum is $(3 - 1) = 2$-shifted symplectic, the associated mapping stack over $\mathsf{T}[1]M$ in four dimensions acquires a canonical (-1)-shifted BV symplectic structure, producing the classical RSVP BV action.

Part II

Lamphrodynamic Field Theory

Chapter 5

The Lamphrodynamic Plenum: Axiomatic Foundation

5.1 Motivation and Overview

The purpose of this chapter is to establish a rigorous axiomatic framework for the lamphrodynamic plenum, consisting of a triple $(\Phi, \vec{v}, S)$ of scalar, vector, and entropy fields. The physical interpretation of these fields has been discussed informally in earlier writings; here we develop a formal foundation suitable for derived geometry and subsequent quantization via the AKSZ construction.

The lamphrodynamic plenum is conceived as a universal medium of information and energetic exchange. The scalar potential Φ encodes lamphrodynamic potential energy, the vector field $\vec{v}$ represents oriented transport flows, and the entropy field S accounts for dissipative and diffusive structure. The fundamental equations couple these fields in a manner that respects derived symplectic geometry, thermodynamic irreversibility, and causal structure.

The classical field theory of $(\Phi, \vec{v}, S)$ will be defined by a variational principle in Section 5.5, producing Euler–Lagrange equations whose analytical properties will be investigated in Chapters 9–11. Our goal in this chapter is to introduce the field variables, state the axioms that govern their behavior, and derive initial differential identities and structural constraints.

5.2 Definition of the Lamphrodynamic Fields

Let (M, g) be a smooth oriented time-oriented Lorentzian manifold of dimension 4. The lamphrodynamic fields consist of:

1. A real-valued scalar field $\Phi \in C^\infty(M, \mathbb{R})$.

2. A spacetime vector field $\vec{v} \in \Gamma(TM)$.

3. A real-valued entropy scalar field $S \in C^\infty(M, \mathbb{R})$.

We write collectively

$$\mathcal{U} := (\Phi, \vec{v}, S).$$

[Field regularity] For the purposes of the classical theory developed in this volume, we assume Φ, S are smooth and $\vec{v}$ is a smooth vector field on M. Weak and distributional extensions will be introduced in Chapter 11.

Remark 5.1. In certain applications we will regard S as taking values in $\mathbb{R}_{\geq 0}$, though for the mathematical theory it suffices to treat S as real-valued.

5.3 Lamphrodynamic Coupling Principles

The physical behavior of the scalar, vector, and entropy fields derives from the principle that lamphrodynamics is governed by coupled transport, diffusion, and potential flow. We impose the following axioms.

[Transport–Potential coupling] The vector field $\vec{v}$ transports the scalar potential Φ according to the advection equation

$$\mathcal{L}_{\vec{v}}\Phi = -\kappa_\Phi \Delta_g \Phi + R_\Phi, \tag{5.1}$$

where $\kappa_\Phi \geq 0$ is a diffusion coefficient, Δ_g is the Laplace–Beltrami operator, and R_Φ is a nonlinear lamphrodynamic coupling term defined in Section 5.6.

[Entropy production] The entropy field S satisfies

$$\mathcal{L}_{\vec{v}}S = \kappa_S \Delta_g S + \sigma(\Phi, \vec{v}, S), \tag{5.2}$$

where $\sigma(\Phi, \vec{v}, S) \geq 0$ is the entropy production rate.

[Velocity response] The transport field $\vec{v}$ responds to both entropy and potential gradients by

$$\nabla_{\vec{v}}\vec{v} = -\nabla\Phi + \beta\nabla S + R_v, \tag{5.3}$$

where $\beta \in \mathbb{R}$ couples entropy gradients to lamphrodynamic flows and R_v is described in Section 5.7.

These axioms are formal for now; precise variational origins will be developed in Section 5.5. The choice of signs corresponds to the interpretation of $-\nabla\Phi$ as attractive potential flow and $\beta\nabla S$ as repulsive entropic response.

5.4 Lamphrodynamic Conservation and Balance Laws

We now derive several differential identities and balance laws implied by the axioms of the previous section. For clarity, we work in local coordinates with the Levi–Civita connection of the Lorentzian metric g.

Proposition 5.2 (Entropy balance)**.** *Assume equation* (5.2)*. Then*

$$\partial_t \int_{\Sigma_t} S \, d\mu_t = \int_{\Sigma_t} \left(\kappa_S \Delta_g S + \sigma(\Phi, \vec{v}, S)\right) d\mu_t \tag{5.4}$$

for any spacelike hypersurface Σ_t with induced measure $d\mu_t$.

Proof. Apply the Reynolds transport theorem to (5.2) and use $\mathcal{L}_{\vec{v}}S = \vec{v}(S)$. □

Similarly, we have a potential energy balance:

Proposition 5.3 (Potential balance)**.** *If* Φ *satisfies* (5.1)*, then*

$$\partial_t \int_{\Sigma_t} \Phi \, d\mu_t = \int_{\Sigma_t} (-\kappa_\Phi \Delta_g \Phi + R_\Phi) \, d\mu_t. \tag{5.5}$$

The vector field $\vec{v}$ obeys a momentum-like balance obtained by integrating (5.3) against $\vec{v}$ and using the identity $\frac{1}{2}\nabla_{\vec{v}}\|\vec{v}\|^2 = \langle \nabla_{\vec{v}}\vec{v}, \vec{v}\rangle$.

Proposition 5.4 (Lamphrodynamic momentum balance)**.** *Assuming* (5.3)*,*

$$\frac{1}{2}\mathcal{L}_{\vec{v}}\|\vec{v}\|^2 = -\langle \nabla\Phi, \vec{v}\rangle + \beta\langle \nabla S, \vec{v}\rangle + \langle R_v, \vec{v}\rangle. \tag{5.6}$$

Remark 5.5. The quantity $\langle \nabla S, \vec{v}\rangle$ plays the role of entropic tension, corresponding physically to emergent repulsion induced by entropy gradients. This term will be crucial in the cosmological applications of Volume II.

5.5 Variational Principle and Euler–Lagrange Equations

We now give the variational origin of the lamphrodynamic equations. Let $\mathcal{L}$ be a Lagrangian density on (M, g) of the form

$$\mathcal{L}(\Phi, \vec{v}, S; g) = \tfrac{1}{2}\|\vec{v}\|^2 + U(\Phi) + W(S) + \alpha\langle \nabla S, \vec{v}\rangle + \beta\langle \nabla\Phi, \vec{v}\rangle, \tag{5.7}$$

where U and W are smooth potential functions and $\alpha, \beta \in \mathbb{R}$ are coupling constants. Let

$$\mathcal{S}[\Phi, \vec{v}, S] := \int_M \mathcal{L} \, dV_g$$

be the associated action functional. Variation against Φ, S, and $\vec{v}$ produces Euler–Lagrange equations which we now compute formally.

Variation with respect to Φ. Integrating by parts yields

$$\delta_\Phi \mathcal{S} = \int_M \big(U'(\Phi) + \beta \operatorname{div}(\vec{v})\big) \, \delta\Phi \, dV_g,$$

and hence the Euler–Lagrange equation is

$$U'(\Phi) + \beta \operatorname{div}(\vec{v}) = 0. \tag{5.8}$$

Variation with respect to S. Similarly,

$$\delta_S \mathcal{S} = \int_M \big(W'(S) + \alpha \operatorname{div}(\vec{v})\big) \, \delta S \, dV_g,$$

producing

$$W'(S) + \alpha \operatorname{div}(\vec{v}) = 0. \tag{5.9}$$

Variation with respect to $\vec{v}$. A computation using the Levi–Civita connection gives

$$\delta_{\vec{v}}\mathcal{S} = \int_M (\vec{v} + \alpha\nabla S + \beta\nabla\Phi)\cdot\delta\vec{v}\,dV_g,$$

leading to

$$\vec{v} + \alpha\nabla S + \beta\nabla\Phi = 0. \tag{5.10}$$

Remark 5.6. Equations (8.2), (8.4), and (8.3) represent the stationary points of the lamphrodynamic action. Their consistency with the axioms of Section 5.3 requires suitable choices of U, W, α, β; Section 5.6 analyzes this structure.

5.6 Nonlinear Couplings and Constraint Relations

We now discuss the nonlinear terms R_Φ and R_v introduced formally in Section 5.3. These terms encode constraint relations among Φ, $\vec{v}$, and S that are not captured by the minimal action (5.7). In physical terms, R_Φ and R_v represent lamphrodynamic self-interaction and dissipation beyond the lowest-order coupling.

[Lamphrodynamic constraint] There exist smooth functions $\mathcal{C}_\Phi(\Phi, \vec{v}, S)$ and $\mathcal{C}_v(\Phi, \vec{v}, S)$ such that

$$R_\Phi = \mathcal{C}_\Phi(\Phi, \vec{v}, S), \qquad R_v = \mathcal{C}_v(\Phi, \vec{v}, S), \tag{5.11}$$

subject to the compatibility condition

$$\operatorname{div}(R_v) + \langle\nabla S, R_v\rangle + \langle\nabla\Phi, R_v\rangle = \Delta_g R_\Phi. \tag{5.12}$$

Remark 5.7. Condition (5.12) guarantees that Lamphrodynamic balance laws are preserved at nonlinear order, and plays a role analogous to constitutive relations in hydrodynamics.

5.7 Symmetries and Noether Currents

We discuss the symmetries of the lamphrodynamic plenum at the level of the classical action (5.7).

Definition 5.8. A local symmetry of the lamphrodynamic action is a variation $(\delta\Phi, \delta\vec{v}, \delta S)$ such that $\delta\mathcal{S} = 0$ for all field configurations satisfying appropriate boundary conditions.

Proposition 5.9 (Shift symmetry of Φ)**.** *If U depends only on $\nabla\Phi$ (e.g. pure gradient theory), then the transformation $\Phi \mapsto \Phi + \epsilon$ is a symmetry of $\mathcal{S}$, yielding a conserved current*

$$J^\mu_\Phi = \frac{\partial\mathcal{L}}{\partial(\nabla_\mu\Phi)}.$$

Remark 5.10. In cosmological settings, U will typically depend on Φ itself, breaking shift symmetry and producing effective potential-induced flow.

Likewise, if W depends only on ∇S, then $S \mapsto S + \epsilon$ is a symmetry; otherwise, entropy production breaks S-shift symmetry. The behavior of these symmetries in the presence of R_Φ and R_v is subtle and will be discussed further in Volume II.

5.8 Conservation Laws and Energy Conditions

We conclude this chapter with an analysis of energy conditions for lamphrodynamic flows. Let

$$\mathcal{E} := \tfrac{1}{2}\|\vec{v}\|^2 + U(\Phi) + W(S)$$

denote the lamphrodynamic energy density. Assuming equations (8.2)–(8.3), one computes

Proposition 5.11 (Lamphrodynamic energy balance)**.**

$$\mathcal{L}_{\vec{v}}\mathcal{E} = -\alpha\|\nabla S\|^2 - \beta\|\nabla\Phi\|^2 + \langle R_v, \vec{v}\rangle + R_\Phi. \tag{5.13}$$

Proof. Differentiating $\mathcal{E}$ along the flow of $\vec{v}$ and substituting (8.2)–(8.3) yields the stated identity. □

Remark 5.12. If $\alpha, \beta > 0$ and $R_\Phi = R_v = 0$, then $\mathcal{L}_{\vec{v}}\mathcal{E} \leq 0$, indicating lamphrodynamic dissipation. The nonlinear regime will be addressed in Chapter 10 via a priori energy estimates.

Summary and Preview. This chapter has introduced the lamphrodynamic fields, stated axioms for their coupling, developed a variational formulation, and identified initial balance laws and symmetries. In Chapter 6 we construct the derived moduli stack of lamphrodynamic configurations and show that it admits a canonical $(3-1) = 2$-shifted symplectic structure, crucial for the AKSZ quantization developed in Chapter 7.

Chapter 6

Derived Stack Construction for RSVP

6.1 Motivation and Overview

In Chapter 5 the lamphrodynamic fields $(\Phi, \vec{v}, S)$ were introduced as classical smooth fields on a Lorentzian spacetime (M, g). To quantize the theory using the AKSZ formalism, we require a *derived* moduli stack whose classical points correspond to such triples, so that the mapping stack

$$\mathrm{Map}(\mathsf{T}[1]M, \mathrm{Plenum})$$

inherits a (-1)-shifted symplectic structure. This chapter constructs the derived moduli stack Plenum, proves that it is a derived Artin stack locally of finite presentation, describes its tangent complex and obstruction theory, and prepares the ground for the shifted symplectic structure in Chapter 7.

We rely on foundational results of Lurie [**Lurie-DAG**] and Toën–Vezzosi [**ToenVezzosi-HAGII**] on derived algebraic geometry, and on Pantev–Toën–Vaquié–Vezzosi [Pan+13] for the construction of shifted symplectic forms on derived mapping stacks.

6.2 The Derived Moduli Functor

Fix a smooth oriented 4-manifold M with Lorentzian metric g. For a derived affine scheme $\mathrm{Spec}\, A$, define the space of lamphrodynamic fields over A by

$$\mathcal{U}(A) := \{(\Phi, \vec{v}, S) \mid \Phi \in \Gamma(M_A, \mathcal{O}_{M_A}),\ \vec{v} \in \Gamma(M_A, T_{M_A}),\ S \in \Gamma(M_A, \mathcal{O}_{M_A})\},$$

where $M_A := M \times \mathrm{Spec}\, A$ and T_{M_A} is the relative tangent complex. We regard $\mathcal{U}$ as a functor

$$\mathcal{U} \colon \mathbf{dAff}^{\mathrm{op}} \longrightarrow \mathbf{sSet}.$$

Definition 6.1. The *derived lamphrodynamic moduli stack* is the derived stackification of $\mathcal{U}$, denoted

$$\mathrm{Plenum} := \mathbf{L}\mathcal{U}.$$

Classical field configurations correspond to $A = \mathbb{R}$, so that Plenum($\mathbb{R}$) recovers the classical triples introduced in Chapter 5.

6.3 Representability and Derived Artin Property

We now establish that Plenum is a derived Artin stack. Let us outline the representability argument; complete details follow the general pattern of mapping stacks of sections of a vector bundle, together with the derived enhancement required by the lamphrodynamic constraint.

Theorem 6.2. *The derived lamphrodynamic moduli stack* Plenum *is a derived Artin stack locally of finite presentation over* $\mathbb{R}$*.*

Proof (sketch). The moduli problem is locally given by sections Φ, S of trivial line bundles and a section $\vec{v}$ of the tangent bundle. Such mapping stacks are representable by derived Artin stacks ([**ToenVezzosi-HAGII**], §7). The nonlinear constraints (5.11)–(5.12) are encoded as derived intersections of smooth Artin stacks, preserving Artin representability. For details see [**Lurie-DAG**; **ToenVezzosi-HAGII**]. □

Remark 6.3. The lamphrodynamic constraint creates a derived structure: classically one would impose $\mathcal{C}_\Phi = \mathcal{C}_v = 0$, but this would produce singularities in the moduli space. Derived geometry resolves these singularities by enriching the intersection with higher homotopical data.

6.4 Local Models and the Tangent Complex

Given $\mathcal{U}$ as above, the tangent complex at a point

$$(\Phi_0, \vec{v}_0, S_0) \in \text{Plenum}$$

is given by the derived global sections

$$T_{(\Phi_0,\vec{v}_0,S_0)}\text{Plenum} \simeq \mathbf{R}\Gamma\big(M, \mathcal{O}_M \oplus T_M \oplus \mathcal{O}_M\big). \tag{6.1}$$

When nonlinear constraints are included, the tangent complex becomes the fiber of

$$\mathbf{R}\Gamma(M, \mathcal{O}_M \oplus T_M \oplus \mathcal{O}_M) \longrightarrow \mathbf{R}\Gamma(M, \mathcal{N}),$$

where $\mathcal{N}$ is the derived normal bundle determined by the Jacobian of the constraint functions $\mathcal{C}_\Phi, \mathcal{C}_v$.

Proposition 6.4. *The tangent complex of* Plenum *has perfect amplitude contained in* $[-1, 1]$*.*

Proof (sketch). The tangent complex of sections of a perfect complex has perfect amplitude given by the amplitude of the underlying complex of sheaves. The derived constraint adds a fiber whose amplitude is bounded by 1. Standard arguments apply [**Lurie-DAG**]. □

Remark 6.5. The amplitude $[-1, 1]$ is crucial: it ensures Plenum has a well-defined shifted symplectic structure of degree 2; see Chapter 7.

6.5 Obstruction Theory and Derived Intersections

We now analyze the derived enhancement induced by the nonlinear lamphrodynamic constraints (5.11)–(5.12). Following [**ToenVezzosi-HAGII**], derived algebraic geometry replaces classical intersections of moduli problems with homotopy fiber products. Let $\mathcal{M}_{\Phi,S}$ denote the moduli stack of pairs of functions $(\Phi, S) \in \Gamma(M, \mathcal{O}_M)^2$, and let $\mathcal{M}_v$ denote the moduli stack of $\Gamma(M, T_M)$-valued sections. The lamphrodynamic constraints define classical morphisms

$$\mathcal{M}_{\Phi,S} \times \mathcal{M}_v \longrightarrow \mathcal{N}_{\Phi,v}$$

whose zero locus would be singular in general.

Definition 6.6. The *derived lamphrodynamic intersection* is the homotopy fiber product

$$\text{Plenum} \simeq (\mathcal{M}_{\Phi,S} \times \mathcal{M}_v) \times^{\mathbf{R}}_{\mathcal{N}_{\Phi,v}} *.$$

Remark 6.7. The derived homotopy fiber product retains information lost by classical intersection, encoding obstructions in higher homotopical degrees. This derived structure is responsible for the shifted symplectic geometry we exploit in Chapter 7.

6.6 Derived Normal Bundle and Obstruction Classes

Let $\mathcal{N}_{\Phi,v}$ denote the normal bundle determined by the Jacobian of the lamphrodynamic constraint functions. The derived normal bundle is obtained by taking the homotopy fiber of the Jacobian mapping between perfect complexes, producing a derived vector bundle in degrees $[-1, 0]$. The obstruction theory for Plenum is hence controlled by the derived cohomology of $\mathcal{N}_{\Phi,v}$, localized on M.

Theorem 6.8. *The derived moduli stack* Plenum *admits a perfect obstruction theory in the sense of [***BF97***], induced by the derived normal bundle of the lamphrodynamic constraint.*

Proof (sketch). The lamphrodynamic constraint defines a morphism of derived Artin stacks. The homotopy fiber of the Jacobian defines a perfect obstruction theory, using the general representability results of [**BF97**; **ToenVezzosi-HAGII**]. □

Remark 6.9. Physically, the obstruction theory describes infinitesimal lamphrodynamic defects: perturbations of $(\Phi, \vec{v}, S)$ that are obstructed by the constraint. In Volume II these play a role near entropic singularities.

6.7 Existence of a 2-Shifted Symplectic Structure

We now show that Plenum admits a natural 2-shifted symplectic structure. The general existence theorem in [Pan+13] applies to derived intersections of moduli of sections of perfect complexes when certain orientation data are available. In the present setting, the Lorentzian manifold (M, g) provides the necessary orientation and volume form, and the kinetic and coupling terms in the action provide a natural closed 3-form.

Let Θ be the lamphrodynamic 3-form defined locally by

$$\Theta := \Phi\, dS \wedge \star d\Phi + \beta\, \langle \nabla\Phi, \nabla S\rangle\, dV_g + \gamma\, \langle \nabla S, \vec{v}\rangle\, d(\iota_{\vec{v}} dV_g), \tag{6.2}$$

where $\star$ is the Hodge operator associated to g and γ is a normalizing constant determined below.

Proposition 6.10. *The form Θ extends to a closed 3-form on the derived moduli stack* Plenum*, defining a 2-shifted presymplectic structure.*

Proof (sketch). Each term in (6.2) is closed at the classical level modulo the lamphrodynamic constraint. The derived extension is obtained by pulling back to the derived intersection and verifying closedness in the derived de Rham complex, following [Pan+13]. Full details will be provided in Appendix E. □

6.8 Local Charts and Explicit Expression for ω

We now identify the 2-shifted symplectic form ω as the derived differential of Θ:

$$\omega := d_{\mathrm{dR}}\Theta.$$

Classically, this yields

$$\omega = d\Phi \wedge dS \wedge \star d\Phi + \Phi\, d(dS \wedge \star d\Phi) \tag{6.3}$$
$$+ \beta\, d\big(\langle \nabla\Phi, \nabla S\rangle\, dV_g\big) + \gamma\, d\big(\langle \nabla S, \vec{v}\rangle\, d(\iota_{\vec{v}} dV_g)\big). \tag{6.4}$$

The derived extension incorporates infinitesimal deformations and ghosts of each lamphrodynamic field. We omit the full expression here, as it requires the total derived de Rham complex on an Artin stack; see Appendix E.

Theorem 6.11 (Shifted symplectic structure of Plenum)**.** *The closed derived 3-form Θ induces a 2-shifted symplectic structure ω on* Plenum*.*

Proof (sketch). Closedness is by construction. Non-degeneracy follows from the perfect amplitude $[-1, 1]$ of the tangent complex and the existence of the lamphrodynamic constraint which produces a derived intersection of section spaces satisfying the orientation axioms of [Pan+13]. See also [**ToenVezzosi-HAGII**]. □

Remark 6.12 (Physical origin of the shift)*.* Since the physical spacetime dimension is 4, integration of the derived 3-form over the source M produces an effective shift by $3 - 4 = -1$ in the BV symplectic degree, giving a (-1)-shifted BV structure in Chapter 7. Hence the physical degree k for RSVP quantization is forced to be 2 at the classical level, consistent with the AKSZ construction for 4-dimensional theories.

6.9 Main Structural Theorem for the Lamphrodynamic Plenum

We now unify the constructions of the preceding sections into a single structural result, collecting assumptions and conclusions in one place.

Theorem 6.13 (Main Structural Theorem)**.** *Let M be a smooth oriented time-oriented 4-dimensional Lorentzian manifold and let $(\Phi, \vec{v}, S)$ be the lamphrodynamic fields with nonlinear constraints (5.11)–(5.12). Then the following properties hold:*

1. *The moduli problem of lamphrodynamic triples extends to a derived functor $\mathcal{U} \colon \mathbf{dAff}^{\mathrm{op}} \to \mathbf{sSet}$ whose derived stackification* Plenum *is a derived Artin stack locally of finite presentation.*

2. *The nonlinear constraints are resolved by a perfect obstruction theory induced from the derived normal bundle of the constraint, and* Plenum *has a tangent complex of perfect amplitude $[-1, 1]$.*

3. *There exists a canonical closed derived 3-form Θ whose derived de Rham differential $\omega := d_{\mathrm{dR}}\Theta$ defines a 2-shifted symplectic structure on* Plenum*.*

4. *The shifted symplectic structure is functorial with respect to pullback along smooth morphisms of Lorentzian spacetimes, and admits local presentations on Artin atlases of* Plenum *compatible with the lamphrodynamic constraint.*

Proof (sketch). Part (1) follows from representability of mapping stacks of sections of perfect complexes and stability of Artin representability under derived homotopy fiber products [**ToenVezzosi-HAGII**]. Part (2) follows from the construction of the derived normal bundle and Behrend–Fantechi obstruction theories [**BF97**]. Part (3) is an instance of the PTVV construction of shifted symplectic forms on derived mapping stacks with perfect obstruction theory [Pan+13]. Part (4) follows from standard functoriality properties of shifted symplectic geometry [**Calaque16**]. □

6.10 Physical Interpretation

Although the shift by 2 appears purely technical, it reflects the fact that RSVP is a 4-dimensional physical theory whose classical degrees of freedom arise from a 3-form Θ. The descent from $k = 2$ to $k - 4 = -2$ under integration over the source would ordinarily yield a (-2)-shift, but the AKSZ construction uses $\mathsf{T}[1]M$ in place of M, producing the canonical (-1)-shift required for the BV bracket. Thus the appearance of degree 2 is a physical consequence of the lamphrodynamic structure in four dimensions.

Remark 6.14. One may view the lamphrodynamic plenum as a universal ambient space of physical fields in which gravitational and entropic interactions are encoded as derived symplectic geometry. In this sense, $(\Phi, \vec{v}, S)$ constitute a pre-geometric substrate from which spacetime causality emerges (developed further in Volume II).

6.11 Preview of the AKSZ–RSVP Action

In Chapter 7 we construct the AKSZ action associated to the 2-shifted symplectic derived stack Plenum. The result will be a (-1)-shifted symplectic BV theory on the mapping stack

$\mathrm{Map}(\mathsf{T}[1]M, \mathrm{Plenum})$, providing a geometric and functorial route to quantization. The Euler–Lagrange equations of the AKSZ action reproduce the lamphrodynamic field equations of Chapter 5, up to additional ghost and antifield terms required for BV consistency.

Remark 6.15. In the AKSZ formalism, no gauge-fixing or ad-hoc ghosts are introduced: the ghost and antifield sectors appear automatically, determined by the shifted symplectic geometry. This is essential for a physically well-defined quantization of RSVP.

Chapter 7

AKSZ–RSVP Action and Classical Master Equation

7.1 Overview and Strategy

The goal of this chapter is to apply the AKSZ construction of Chapter 4 to the derived lamphrodynamic plenum constructed in Chapter 6. The result is a (-1)-shifted symplectic BV theory whose classical Euler–Lagrange equations agree with the lamphrodynamic equations of Chapter 5. This constitutes the precise mathematical definition of the RSVP field theory at the classical level.

Let (M, g) be a smooth oriented time-oriented 4-dimensional Lorentzian manifold with associated shifted tangent bundle $\mathsf{T}[1]M$. Define the AKSZ field space

$$\mathcal{F} := \mathrm{Map}(\mathsf{T}[1]M, \mathrm{Plenum}),$$

a derived mapping stack equipped with the canonical (-1)-shifted symplectic form of Chapter 4. Fields in $\mathcal{F}$ consist of superfields whose degree-zero components are the lamphrodynamic fields $(\Phi, \vec{v}, S)$ and whose higher-degree components constitute ghosts, antifields, and higher ghosts.

7.2 AKSZ Data for RSVP

The AKSZ construction requires:

1. a source $(\mathsf{T}[1]M, d_M)$ with homological vector field d_M, the de Rham differential on M shifted by 1;

2. a target $(\mathrm{Plenum}, \omega)$ with 2-shifted symplectic form ω and homological vector field Q_{Plenum} preserving ω;

3. a Hamiltonian function α on Plenum satisfying $Q_{\mathrm{Plenum}} = \{\alpha, -\}$.

Under these assumptions, the AKSZ recipe produces the BV action

$$S_{\mathrm{AKSZ}} = \int_{\mathsf{T}[1]M} \langle \Phi^* \theta, d_M \Phi \rangle + \Phi^* \alpha,$$

where θ is a potential for the shifted symplectic form on Plenum. Explicit formulas for θ will be given below (Section 7.4).

7.3 Derived 3-Form and BV Potential

Recall from Chapter 6 the derived 3-form Θ on Plenum given locally by

$$\Theta = \Phi\, dS \wedge \star d\Phi + \beta\, \langle \nabla\Phi, \nabla S\rangle\, dV_g + \gamma\, \langle \nabla S, \vec{v}\rangle\, d(\iota_{\vec{v}} dV_g).$$

Let θ be any degree-2 potential for $\omega = d_{\mathrm{dR}}\Theta$ in the derived de Rham complex of Plenum. Such a potential exists locally and may be constructed globally by descent, using the Artin atlases of Plenum.

Remark 7.1. In practice, θ may be chosen to be the 2-form part of Θ, omitting terms that vanish under pullback to the mapping stack. However, the existence of a global potential is essential for defining the AKSZ action.

7.4 The RSVP AKSZ Action

The AKSZ action is now defined by

$$S_{\mathrm{RSVP}} := \int_{\mathsf{T}[1]M} \left(\langle \Phi^*\theta, d_M\Phi\rangle + \Phi^*\alpha \right). \tag{7.1}$$

In local coordinates, writing $\Phi^* = (\Phi, \vec{v}, S)$ for the pullback of the universal fields, we may express the integrand as

$$\mathcal{L}_{\mathrm{AKSZ}} = \langle d_M\Phi, \theta_\Phi\rangle + \langle d_M\vec{v}, \theta_v\rangle + \langle d_M S, \theta_S\rangle + \alpha(\Phi, \vec{v}, S), \tag{7.2}$$

where $\theta_\Phi, \theta_v, \theta_S$ are the components of the BV potential paired with the corresponding superfield derivatives.

Proposition 7.2. *The functional S_{RSVP} is well defined on the derived mapping stack $\mathcal{F}$ and has total ghost number zero.*

Proof (sketch). Total degree zero follows from degree counting: the degree of d_M is $+1$, the degree of θ is 2, and the integration over $\mathsf{T}[1]M$ reduces degree by 3, yielding degree 0 overall. See [**CF07**; Ale+97]. □

7.5 Classical Master Equation

Let $\{-,-\}_{\mathrm{BV}}$ denote the BV antibracket induced by the (-1)shifted symplectic structure on $\mathcal{F} = \mathrm{Map}(\mathsf{T}[1]M, \mathrm{Plenum})$. The classical master equation is

$$\{S_{\mathrm{RSVP}}, S_{\mathrm{RSVP}}\}_{\mathrm{BV}} = 0.$$

Theorem 7.3 (Classical Master Equation)**.** *The RSVP AKSZ action* (7.1) *satisfies the classical master equation*

$$\{S_{\mathrm{RSVP}}, S_{\mathrm{RSVP}}\}_{\mathrm{BV}} = 0,$$

and hence defines a classical BV theory.

Proof (Sketch). This follows from Theorem 6.13 and the general AKSZ theorem of AlexandrovKontsevichSchwarzZaboronsky: whenever the target carries a $k$shifted symplectic structure with Hamiltonian $Q_{\mathrm{Plenum}} = \{\alpha, -\}$, the induced action on $\mathrm{Map}(\mathsf{T}[1]M, \mathrm{Plenum})$ satisfies the classical master equation and induces the correct BV differential $Q = \{S, -\}$. □

Remark 7.4. Note that no gauge-fixing is used. The ghosts and antifields arise canonically from the graded geometry of $\mathrm{Map}(\mathsf{T}[1]M, \mathrm{Plenum})$ and the shifted symplectic structure.

7.6 BV Operator and Field Content

The BV differential is defined by

$$Q_{\mathrm{BV}}(F) := \{S_{\mathrm{RSVP}}, F\}_{\mathrm{BV}}, \qquad F \in \mathcal{O}(\mathcal{F}).$$

The field content consists of the superfields

$$\mathbf{\Phi} = \Phi + \Phi^{+} + \Phi^{*} + \cdots, \qquad \mathbf{v} = \vec{v} + v^{+} + v^{*} + \cdots, \qquad \mathbf{S} = S + S^{+} + S^{*} + \cdots,$$

with ghost number assignments determined by the degree shifts of $\mathsf{T}[1]M$ and Plenum. The precise assignments follow the standard AKSZ convention:

$$\mathrm{gh}(\text{field}) = 0, \qquad \mathrm{gh}(\text{ghost}) = +1, \qquad \mathrm{gh}(\text{antifield}) = -1, \qquad \mathrm{gh}(\text{higher ghosts}) > +1.$$

Proposition 7.5. *The BV operator Q_{BV} is a degree $+1$ derivation satisfying $Q_{\mathrm{BV}}^2 = 0$.*

Proof (Sketch). $Q_{\mathrm{BV}}^2(F) = \{S, \{S, F\}\} = \frac{1}{2}\{\{S, S\}, F\}$, which vanishes by Theorem 7.3. □

7.7 EulerLagrange Equations and Lamphrodynamics

We now extract the classical physical equations of motion as the EulerLagrange equations of the AKSZ action at ghost number zero.

Theorem 7.6 (RSVP Equations from AKSZ)**.** *Let S_{RSVP} be the AKSZ action* (7.1)*. Then the ghost-number-zero Euler–Lagrange equations are equivalent to the lamphrodynamic field equations of Chapter 5: namely,*

$$\Box\Phi = \lambda\nabla\cdot\vec{v} + \ldots, \qquad \nabla\cdot\vec{v} = f(\nabla\Phi, \nabla S), \qquad \partial_t S + \nabla\cdot(S\vec{v}) = \kappa\nabla^2 S + \ldots,$$

with all nonlinear source terms determined by the same derived 3form Θ used in the AKSZ construction.

Proof (Sketch). The variation of S_{RSVP} with respect to $(\Phi, \vec{v}, S)$ at ghost number zero yields the classical field equations. All antifield and ghost terms vanish at ghost number zero, leaving precisely the lamphrodynamic PDE system of Chapter 5. □

Remark 7.7. Thus the AKSZ construction is not merely compatible with lamphrodynamics; it *determines* the lamphrodynamic equations as the classical sector of a (-1)shifted symplectic BV theory. The derived geometry forces the PDE system rather than assuming it.

7.8 Physical Interpretation

The significance of the AKSZ construction is that it endows RSVP with a canonical BV quantization framework without requiring gauge-fixing or ad-hoc ghost terms: the ghost sector is automatic. Moreover, the classical lamphrodynamic dynamics are precisely the Euler–Lagrange equations of the BV action, making RSVP a field theory in the strongest modern sense. Finally, the (-1)shift ensures that the BV bracket is well-defined and that the master equation holds, a prerequisite for any rigorous quantum theory built on the BV formalism.

7.9 Boundary Data and BV–BFV Structures

On a manifold with boundary $\partial M \neq \emptyset$, the AKSZ construction produces a canonical BV–BFV structure in the sense of Cattaneo–Mnev–Reshetikhin. Concretely, the boundary data consist of a BFV space of boundary fields $\mathcal{F}_\partial$ and a degree 0 *BFV action* S_∂ which satisfies

$$Q_{\mathrm{BFV}}^2 = 0, \qquad Q_{\mathrm{BFV}}(S_\partial) = 0,$$

and whose cohomology governs physical boundary degrees of freedom.

Theorem 7.8 (Canonical BV–BFV Boundary Structure)**.** *Let M be a 4–manifold with (possibly empty) boundary. Then the AKSZ action S_{RSVP} induces a canonical BV–BFV structure $(\mathcal{F}_\partial, S_\partial)$ on ∂M. Moreover, S_∂ determines the physical boundary excitations of the lamphrodynamic fields $(\Phi, \vec{v}, S)$, including entropic flux across ∂M.*

Proof (Sketch). Immediate from the general BV–BFV theorem for AKSZ models, using that the target is (-1)–shifted symplectic and that M has dimension 4. The boundary inherits a degree-0 Hamiltonian system governing the physical boundary data. □

Remark 7.9 (Physical Meaning)*.* The presence of a canonical BV–BFV boundary theory implies that entropic flux and vector circulation admit well-defined “boundary” degrees of freedom even before gauge-fixing. This suggests a direct route to black-hole and horizon physics once appropriate boundary conditions are chosen (Volume II).

7.10 Local Coordinates and Expansion

Let $(x^\mu)_{\mu=0}^3$ be local coordinates on M and let $(\Phi, \vec{v}, S)$ denote the degree-0 components of the superfields. Expanding the AKSZ action to leading (classical) order gives

$$S_{\mathrm{RSVP}} \simeq \int_M \Big(\partial_\mu \Phi \, \Theta^{\mu\nu\rho} \, \partial_\nu v_\rho + \partial_\mu S \, \Theta^{\mu\nu\rho} \, \partial_\nu \Phi + \dots \Big) \, d^4x, \tag{7.3}$$

where $\Theta^{\mu\nu\rho}$ are the components of the closed 3–form representing the (-1)–shifted symplectic structure (cf. Theorem 6.13).

Remark 7.10. Subleading terms encode the nonlinear couplings responsible for entropy flow and lamphrodynamic vorticity. These terms become crucial in the strong-flux regime and near entropic singularities (Ch. 12).

7.11 Comparison to Chern–Simons AKSZ

A useful analogy is provided by 3–dimensional Chern–Simons theory, which is the prototypical AKSZ model in dimension 3 with target $\mathfrak{g}[1]$.

7.11.1 Chern–Simons AKSZ recap

Let N be a 3–manifold and $\mathfrak{g}$ a Lie algebra. The AKSZ construction gives the classical CS action

$$S_{\mathrm{CS}} = \int_N \Big(A \wedge dA + \tfrac{1}{3} A \wedge [A, A]\Big),$$

arising from the canonical symplectic structure on $\mathfrak{g}[1]$ and the Chevalley–Eilenberg differential.

7.11.2 Structural analogy

Two structural parallels are immediate:

1. **Degree shift.** CS uses a 0–shifted target $\mathfrak{g}[1]$, whereas RSVP uses a (-1)–shifted derived stack. The lower shift reflects the presence of physical scalar and vector fields rather than a pure gauge sector.

2. **Closed form generating the action.** In both theories, the action is generated from the geometric data of the target: the Lie algebra $(\mathfrak{g}, [-, -])$ in CS, the lamphrodynamic 3–form Θ in RSVP.

7.11.3 Physical meaning

For Chern–Simons, the AKSZ construction explains why the CS action is automatically topological and why its BVBFV boundary theory captures edge modes (e.g. quantum Hall). In RSVP the AKSZ construction explains why lamphrodynamic equations arise as Euler–Lagrange equations of a (-1)–shifted BV theory, and thus why entropy flow and vector circulation are *geometric* rather than phenomenological.

Remark 7.11. The fact that lamphrodynamics sits one degree below the CS example suggests a natural comparison between topological edge modes and entropy-flow boundary modes. This will reappear in Chapter 12 (entropic singularities) and Volume II (black-hole analogs).

7.12 Summary and Forward Pointer

In this chapter we have:

1. Constructed the AKSZ action for RSVP from the (-1)–shifted target Plenum.
2. Verified the classical master equation.
3. Identified the BV operator and ghost/antifield expansion.
4. Derived lamphrodynamic PDEs as Euler–Lagrange equations.
5. Described canonical boundary BFV structure.
6. Compared RSVP to the Chern–Simons AKSZ example.

In the next chapter we develop the variational and Hamiltonian formulation of RSVP in classical degree zero, proving that the (-1)–shifted symplectic geometry induces a well-defined Hamiltonian system and enabling energy estimates and Noether-type conservation laws. This prepares the ground for the analytic theory developed in Chapters 9–11.

Chapter 8

Variational Principles and Hamiltonian Formulation

The AKSZ construction of the previous chapter provides a BV master action S_{RSVP} on the graded space of superfields. In this chapter we extract the classical (degree 0) variational principle, identify the induced Euler–Lagrange equations in ordinary fields $(\Phi, \vec{v}, S)$, and construct the classical Hamiltonian formulation on the derived mapping stack. This leads to a well-defined notion of conserved currents and prepares the ground for the analytic theory developed in Chapters 9–11.

8.1 The Classical Variational Functional

Let M be a smooth, oriented 4–manifold without boundary (the boundary case will be handled in the BV–BFV formalism). Restrict the AKSZ action S_{RSVP} to the classical (ghost number 0) sector. Denote by

$$(\Phi, \vec{v}, S) \in \Gamma(M, \mathcal{E})$$

the physical lamphrodynamic fields. Then the classical action takes the form

$$\mathcal{S}[\Phi, \vec{v}, S] = \int_M \mathcal{L}(\Phi, \vec{v}, S, \partial\Phi, \partial\vec{v}, \partial S)\, d^4x, \tag{8.1}$$

where the Lagrangian density $\mathcal{L}$ is obtained by truncating the AKSZ superfield expansion and selecting the degree 0 component (including nonlinear terms arising from the (-1)–shifted symplectic structure).

Remark 8.1. The explicit coordinate expression of $\mathcal{L}$ is cumbersome due to the nonlinear couplings between $\Phi, \vec{v}$ and S. However, the structural properties—gauge invariances, conservation laws, and entropy-flux terms—follow from the (-1)–shifted geometry rather than from the coordinate expansion.

8.2 Euler–Lagrange Equations

The Euler–Lagrange equations are obtained by taking first variations of (8.1):

$$\frac{\delta \mathcal{S}}{\delta \Phi} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi)} \right) - \frac{\partial \mathcal{L}}{\partial \Phi} = 0, \tag{8.2}$$

$$\frac{\delta \mathcal{S}}{\delta v_\nu} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu v_\nu)} \right) - \frac{\partial \mathcal{L}}{\partial v_\nu} = 0, \tag{8.3}$$

$$\frac{\delta \mathcal{S}}{\delta S} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu S)} \right) - \frac{\partial \mathcal{L}}{\partial S} = 0. \tag{8.4}$$

Theorem 8.2 (Equivalence with AKSZ Euler–Lagrange Equations)**.** *The equations* (8.2)–(8.4) *coincide with the classical Euler–Lagrange equations obtained from the AKSZ action* S_{RSVP} *by projecting to degree* 0*.*

Proof. The AKSZ construction guarantees that the BV Euler–Lagrange operator $\delta S_{\mathrm{RSVP}}/\delta \mathsf{Field} = 0$ restricts to the classical sector by ghost number. Truncating to ghost number 0 and using the fact that the BV bracket reduces to the classical Poisson bracket on physical fields yields (8.2)–(8.4). □

8.3 Hamiltonian Formulation on the Derived Mapping Stack

Let $\mathrm{Map}(M, \mathrm{Plenum})$ be the classical mapping space of M into the derived target Plenum, and let $\omega_{(-1)}$ denote the (-1)–shifted symplectic form on Plenum. In the AKSZ formalism, the Hamiltonian structure on the space of superfields descends (after truncation) to a classical Hamiltonian system on $\mathrm{Map}(M, \mathrm{Plenum})$ endowed with the induced (shifted) 2–form.

Proposition 8.3 (Classical Hamiltonian Structure)**.** *The classical space of fields* $\mathrm{Map}(M, \mathrm{Plenum})$ *carries a well-defined Hamiltonian structure* (ω, H) *where*

$$H = \int_M \mathcal{H}(\Phi, \vec{v}, S, \partial\Phi, \partial\vec{v}, \partial S)\, d^4x,$$

and the Hamiltonian density $\mathcal{H}$ *is obtained by classical Legendre transform of the Lagrangian density* $\mathcal{L}$*.*

Proof (Sketch). The (-1)–shifted symplectic form on Plenum induces a 0–shifted (classical) symplectic form on the mapping space. The Legendre transform is defined fiberwise on the classical field bundle $\mathcal{E} \to M$. Standard AKSZ arguments show that the resulting Hamiltonian generates the same equations of motion as (8.2)–(8.4). □

Remark 8.4 (Odd vs Even Poisson Brackets)*.* The BV bracket on the AKSZ superfields is odd (degree $+1$), whereas the induced bracket on the classical fields is even. This provides a natural separation between quantum BV information (ghost/antifield sector) and classical conservation laws.

8.4 Noether Currents and Conserved Quantities

Let $\mathcal{G}$ be a Lie group of symmetries acting on the classical fields $(\Phi, \vec{v}, S)$ via smooth bundle automorphisms compatible with the lamphrodynamic constraint. Write $\rho : \mathfrak{g} \to \Gamma(T\mathcal{E})$ for the induced Lie algebra action on fields.

Definition 8.5 (Infinitesimal Symmetry)**.** A vector field $X \in \Gamma(T\mathcal{E})$ is an infinitesimal symmetry if the Lie derivative of the Lagrangian density satisfies

$$\mathcal{L}_X \mathcal{L} = \partial_\mu B^\mu$$

for some current B^μ.

Standard variational arguments yield:

Proposition 8.6 (Noether Current)**.** *Let X be an infinitesimal symmetry associated with $\xi \in \mathfrak{g}$. Then the Noether current is*

$$J^\mu_\xi = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi)} X(\Phi) + \frac{\partial \mathcal{L}}{\partial(\partial_\mu v_\nu)} X(v_\nu) + \frac{\partial \mathcal{L}}{\partial(\partial_\mu S)} X(S) - B^\mu,$$

and is conserved along classical solutions: $\partial_\mu J^\mu_\xi = 0$.

Remark 8.7. In the BV formulation of Chapter 7, the same current arises as the degree–0 component of the BV charge associated to the ghost–ξ. Hence classical conservation laws are shadows of BV identities.

8.5 Hamiltonian Vector Fields and Poisson Structure

Let ω denote the classical symplectic form on $\mathrm{Map}(M, \mathrm{Plenum})$. For any classical observable $\mathcal{O}$, define the Hamiltonian vector field $X_\mathcal{O}$ by the relation

$$\iota_{X_\mathcal{O}} \omega = \delta \mathcal{O}.$$

Provided ω is weakly nondegenerate on the classical sector, this determines $X_\mathcal{O}$ formally and induces a Poisson bracket

$$\{\mathcal{O}_1, \mathcal{O}_2\} := X_{\mathcal{O}_1}(\mathcal{O}_2),$$

which restricts to field polynomials under classical truncation.

Remark 8.8. Weak nondegeneracy follows from nondegeneracy of the (-1)–shifted symplectic form on the target stack Plenum and the AKSZ descent; a detailed discussion will appear in Appendix E.

8.6 Energy, Momentum, and Entropic Contributions

Energy–momentum for lamphrodynamic fields requires special care because S contributes an entropic stress–energy component not present in standard relativistic field theories.

Definition 8.9 (Lamphrodynamic Stress–Energy Tensor)**.** Define the symmetric tensor

$$T_{\mu\nu} := \frac{\partial \mathcal{L}}{\partial(\partial^\mu \Phi)}\partial_\nu \Phi + \frac{\partial \mathcal{L}}{\partial(\partial^\mu v_\rho)}\partial_\nu v_\rho + \frac{\partial \mathcal{L}}{\partial(\partial^\mu S)}\partial_\nu S - g_{\mu\nu}\mathcal{L},$$

where $g_{\mu\nu}$ is the background Lorentzian metric on M.

Proposition 8.10 (Energy Balance Identity)**.** *Along classical solutions of the Euler–Lagrange equations* (8.2)*–*(8.4)*, the stress–energy tensor satisfies*

$$\nabla^\mu T_{\mu\nu} = \mathscr{E}_\nu(\Phi, \vec{v}, S),$$

where the entropic source term $\mathscr{E}_\nu$ vanishes for isentropic flows and encodes entropy–flux contributions in general.

Remark 8.11. The presence of $\mathscr{E}_\nu$ is physically crucial: it generates effective geometric backreaction in emergent gravity (Volume II, Chapter 1).

8.7 Hamiltonian Energy Estimates and PDE Well–Posedness

The classical Hamiltonian $H = \int_M \mathcal{H}\, d^4x$ provides the fundamental energy control for the analytic theory of Chapters 9–11. The following preparatory result connects the formal Hamiltonian with Sobolev control.

Theorem 8.12 (Hamiltonian Energy Estimate)**.** *Let $(\Phi, \vec{v}, S)$ be a classical solution of the lamphrodynamic field equations on a globally hyperbolic spacetime (M, g). Assume initial data in H^k, $k \geq 2$, and suppose the entropic term satisfies a coercivity condition $\mathscr{E}_0 \leq C|\nabla S|^2$. Then for sufficiently small time $T > 0$,*

$$\sup_{t\in[0,T]} \left(\|\Phi(t)\|_{H^k} + \|\vec{v}(t)\|_{H^k} + \|S(t)\|_{H^k}\right) \;\leq\; C_0 \exp(CT) \left(\|\Phi_0\|_{H^k} + \|\vec{v}_0\|_{H^k} + \|S_0\|_{H^k}\right),$$

where C_0, C depend on the geometry of M.

Proof (Sketch). The Hamiltonian controls the H^k–norms modulo entropic terms. Coercivity of $\mathscr{E}_0$ and Grönwall's inequality yield the exponential bound. A detailed PDE proof is given in Chapter 10. □

8.8 Summary and Forward Link

In summary, the classical limit of the AKSZ–RSVP theory yields:

1. a well-defined variational principle,
2. Euler–Lagrange equations equivalent to the classical AKSZ equations,
3. a Hamiltonian formalism on the derived mapping stack,
4. Noether currents and entropic stress–energy,

5. energy estimates sufficient for local well-posedness.

These results serve as the bridge between the geometric BV framework (Chapters 4–7) and the analytic theory (Chapters 9–11), where energy estimates and Sobolev bounds play a central role.

Part III

Well-Posedness and Analysis

Chapter 9

Linear Theory

Chapters 5–8 developed the geometric and variational structure of lamphrodynamic fields $(\Phi, \vec{v}, S)$, culminating in the classical Hamiltonian formulation. We now turn to analytic questions, beginning with the linearization of the Euler–Lagrange equations (8.2)–(8.4) about a smooth reference configuration and establishing global well–posedness in Sobolev spaces. This chapter collects the functional–analytic tools needed for the nonlinear theory of Chapters 10–11.

9.1 Linearization about a Reference Configuration

Let $(\Phi, \vec{v}, S)$ be a smooth lamphrodynamic configuration solving the classical Euler–Lagrange equations on a globally hyperbolic spacetime (M, g). Define the linearized fields by

$$(\phi, w_\mu, s) := \frac{d}{d\varepsilon}\Big|_{\varepsilon=0}(\Phi + \varepsilon\phi,\ v_\mu + \varepsilon w_\mu,\ S + \varepsilon s).$$

To simplify notation, write $U := (\phi, w, s)$. Substituting into (8.2)–(8.4) and retaining terms linear in U yields a linear system of wave–type operators

$$\mathcal{L}_{\text{lin}} U = \Box_g U + B^\mu \nabla_\mu U + C\,U = 0, \tag{9.1}$$

where $\Box_g$ is the tensor wave operator, B^μ is a first–order coefficient determined by the background $(\Phi, \vec{v}, S)$, and C is a zero–order coefficient. The coefficients B^μ, C are smooth and bounded on compact sets.

Remark 9.1 (Hyperbolic Character). The principal part of (9.1) is the wave operator, ensuring hyperbolicity; lower–order terms encode entropy–flux corrections of the background.

9.2 Functional Spaces and Initial Value Problem

Fix a spacelike Cauchy hypersurface Σ with induced Riemannian metric h. For integer $k \geq 0$ define Sobolev norms

$$\|U\|_{H^k(\Sigma)} := \sum_{j=0}^{k} \|\nabla^j U\|_{L^2(\Sigma)}.$$

Definition 9.2 (Initial Value Problem)**.** Given initial data

$$U(0) = U_0 \in H^k(\Sigma), \qquad \partial_t U(0) = U_1 \in H^{k-1}(\Sigma),$$

find $U(t, x)$ solving (9.1) for t in some time interval containing 0.

9.3 Energy Estimates

Define the linear energy functional

$$E_{\text{lin}}(U)(t) := \frac{1}{2} \int_{\Sigma_t} \left(|\partial_t U|^2 + |\nabla U|^2 + |U|^2 \right) d\text{vol}_h. \tag{9.2}$$

Standard integration–by–parts yields:

Lemma 9.3 (Linear Energy Estimate)**.** *For sufficiently smooth U solving* (9.1)*,*

$$\frac{d}{dt} E_{\text{lin}}(U)(t) \leq C(E_{\text{lin}}(U)(t) + \|U(t)\|^2_{L^2}),$$

where C depends on the background coefficients.

Proof (Sketch). Multiply (9.1) by $\partial_t U$, integrate over Σ_t, use divergence theorem on the wave operator, and absorb lower–order terms into the right–hand side. □

By Grönwall's inequality we obtain:

Corollary 9.4 (A priori Control)**.** *For $t \in [0, T]$,*

$$E_{\text{lin}}(U)(t) \leq E_{\text{lin}}(U)(0) \exp(CT).$$

9.4 Global Well–posedness

We now state the main linear result.

Theorem 9.5 (Global Well–posedness in Sobolev Spaces)**.** *Let (M, g) be globally hyperbolic and let $k \geq 2$. For any initial data $(U_0, U_1) \in H^k(\Sigma) \times H^{k-1}(\Sigma)$ there exists a unique global solution*

$$U \in C^0([0, \infty); H^k(\Sigma)) \cap C^1([0, \infty); H^{k-1}(\Sigma))$$

of (9.1)*. Moreover, the energy bound of Corollary 9.4 holds.*

Proof. Local existence and uniqueness follow from standard hyperbolic PDE theory for systems with smooth coefficients [**Hormander-AnalysisPDE**]. Global extension follows from the a priori bound (Corollary 9.4) combined with continuation criteria. □

9.5 Spectral Properties of the Linearized Operator

On static spacetimes or after Fourier transform in time, $\mathcal{L}_{\mathrm{lin}}$ reduces to an elliptic operator of the form

$$\mathscr{A} = -\Delta_h + V(x),$$

where V is a potential induced by the background. Standard spectral theory implies:

Proposition 9.6 (Spectral Properties)**.** *If (M, g) is static and Σ compact, then $\mathscr{A}$ has discrete spectrum $\{\lambda_j\}_{j\geq 0}$, with $\lambda_j \to +\infty$ as $j \to \infty$.*

Proof. Immediate from ellipticity and compactness of Σ [**Taylor-PDE**]. □

9.6 Connection to AKSZ Classical Limit

The linear theory developed here governs fluctuations around classical solutions of the AKSZ–RSVP action. The coercivity of the Hamiltonian (Chapter 8) and the linear estimate above ensure that the BV perturbative expansion is formally well behaved around smooth backgrounds, a prerequisite for quantization in Volume III.

9.7 Summary

We have shown that linearized lamphrodynamic fields satisfy a hyperbolic system of second–order PDEs with global well–posedness in Sobolev spaces. The resulting energy and spectral control forms the analytic basis for the nonlinear theory of Chapters 10–11.

Chapter 10

Energy Estimates and Regularity

The global well–posedness of the linearized lamphrodynamic system (Chapter 9) forms the analytic foundation for the nonlinear theory. We now establish higher–order energy estimates for the full nonlinear lamphrodynamic equations, prove local well–posedness in Sobolev spaces, and derive regularity criteria that exert control near potential entropic singularities. These results prepare the ground for the global continuation problem in Chapter 11.

10.1 Nonlinear Lamphrodynamic System

Let $(\Phi, \vec{v}, S)$ satisfy the Euler–Lagrange equations (8.2)–(8.4). Suppressing indices, write the nonlinear system schematically as

$$\Box_g U = \mathcal{N}(U, \nabla U), \tag{10.1}$$

where $U = (\Phi, \vec{v}, S)$ and $\mathcal{N}$ is a smooth nonlinear mapping whose algebraic form depends polynomially on U and its first derivatives; in particular, no derivative loss occurs at the level of principal part.

10.2 Higher–order Energies

For integer $k \geq 2$ define the k–th order energy

$$E_k(t) = \frac{1}{2} \sum_{|\alpha| \leq k} \int_{\Sigma_t} \Big(|\partial_t \nabla^\alpha U|^2 + |\nabla \nabla^\alpha U|^2 + |\nabla^\alpha U|^2 \Big) \, d\mathrm{vol}_h, \tag{10.2}$$

where α is a spatial multi-index and ∇^α denotes the covariant derivative with respect to h.

Our goal is to estimate $E_k(t)$ in terms of initial data $E_k(0)$ and lower–order terms.

Lemma 10.1 (Commutator Estimate)**.** *For each $|\alpha| \leq k$ one has*

$$\big\| [\nabla^\alpha, \mathcal{N}] \, U \big\|_{L^2(\Sigma_t)} \leq C \sum_{j \leq k} \|\nabla^j U\|^2_{L^2(\Sigma_t)}.$$

Proof (Sketch). The nonlinearities are polynomial in U and ∇U with smooth coefficients, so commutators produce lower–order terms which are controlled by standard Moser–type estimates. □

10.3 Nonlinear Energy Inequality

Commuting ∇^α with (10.1), multiplying by $\partial_t \nabla^\alpha U$, integrating over Σ_t, and using Lemma 10.1 yields the fundamental inequality

$$\frac{d}{dt} E_k(t) \leq C(E_k(t) + E_k(t)^{3/2}). \tag{10.3}$$

Remark 10.2. The cubic growth reflects the polynomial nonlinearity of the lamphrodynamic interactions; the crucial point is absence of derivative loss, ensuring that the H^k–norm remains the natural energy scale.

10.4 Local Well–posedness

We now state the main nonlinear local existence theorem.

Theorem 10.3 (Local Well–posedness in H^k)**.** *Let $k \geq 2$ and initial data $U_0 \in H^k(\Sigma)$, $U_1 \in H^{k-1}(\Sigma)$. Then there exists $T > 0$ and a unique solution*

$$U \in C^0([0,T]; H^k(\Sigma)) \cap C^1([0,T]; H^{k-1}(\Sigma))$$

of the nonlinear system (10.1)*.*

Proof (Sketch). Iterate the linear problem with nonlinear forcing; apply the energy inequality (10.3) and Grönwalls inequality to obtain uniform bounds on a short time interval. Contraction mapping in the energy norm yields uniqueness. □

10.5 Continuation Criterion and Blowup Scenarios

Let $T_{\max}$ denote the maximal time of existence in the class of H^k–solutions.

Theorem 10.4 (Continuation Criterion)**.** *If $T_{\max} < \infty$, then*

$$\limsup_{t \nearrow T_{\max}} \Big(\|U(t)\|_{H^k} + \|\nabla U(t)\|_{L^\infty} \Big) = +\infty.$$

Proof (Sketch). Suppose the right–hand side remains finite; controlling ∇U in L^∞ yields uniform bounds for nonlinear terms, contradicting maximality. Standard hyperbolic PDE techniques apply. □

Remark 10.5 (Possible Singularities)*.* Entropic singularities may arise precisely when gradients of S become unbounded. Understanding such scenarios is central to the global existence problem (Chapter 11).

10.6 Higher Regularity and Sobolev Embeddings

Assume the solution U lies in H^k for $k > \frac{n}{2} + 1 = 3$ in spatial dimension 3; then Sobolev embedding implies

$$\|U\|_{C^1} \leq C\|U\|_{H^k}.$$

Together with (10.3), this yields:

Corollary 10.6 (Higher Regularity)**.** *If initial data belong to H^k, $k > 3$, then the solution remains C^1 for as long as it exists.* □

10.7 Entropy Coercivity and Stability

Entropy plays a stabilizing role when its contribution satisfies a coercivity condition similar to that in Theorem 8.12. Assume

$$\mathscr{E}_0(U) \leq C|\nabla S|^2. \tag{10.4}$$

Then:

Proposition 10.7 (Stability under Entropy Coercivity)**.** *If* (10.4) *holds, then there exist constants $C_0, C > 0$ such that*

$$E_k(t) \leq C_0 E_k(0) \exp(CT), \qquad 0 \leq t \leq T.$$

Proof (Sketch). Combine (10.4) with (10.3) and apply Grönwalls inequality. □

10.8 Summary and Forward Link

We have proved local well–posedness, higher–order energy estimates, and a continuation criterion for the full nonlinear lamphrodynamic equations. The analysis hinges on coercivity of entropic terms and the absence of derivative loss. In Chapter 11, we study global continuation, entropy–driven singularities, and prospects for global existence.

Chapter 11

Nonlinear Global Theory

The local theory of Chapter 10 guarantees existence and uniqueness of classical lamphrodynamic solutions on a short time interval, subject to finite Sobolev norms. Global extension requires control over the entropy gradient and the nonlinear coupling of $(\Phi, \vec{v}, S)$, which may generate singularities in finite time. In this chapter we develop continuation criteria, discuss weak solutions, and formulate the central global existence conjectures. Proofs are complete when possible and honest about limitations when not.

11.1 Entropy–Driven Singularities

Let $U = (\Phi, \vec{v}, S)$ solve the nonlinear system (10.1). The most severe potential singularities arise from blow–up of the entropy gradient ∇S.

Proposition 11.1 (Entropy Singularity Criterion). *If a classical solution loses smoothness at finite time $T_{\max} < \infty$, then*

$$\limsup_{t \nearrow T_{\max}} \|\nabla S(t)\|_{L^\infty} = +\infty.$$

Proof (Sketch). If $\|\nabla S(t)\|_{L^\infty}$ remained bounded, then by Theorem 10.4, the H^k–norms would also remain bounded, contradicting maximality of $T_{\max}$. □

Remark 11.2. This establishes the entropy gradient as the natural blow–up variable for lamphrodynamic fields, in contrast to standard fluid models where density or vorticity dominate.

11.2 Global Existence under Small Data

On Minkowski space $\mathbb{R}^{1+3}$, global existence can be shown for sufficiently small initial data by exploiting the dispersive character of the wave operator.

Theorem 11.3 (Small–Data Global Existence). *Let $M = \mathbb{R}^{1+3}$ with flat metric and assume $\|U_0\|_{H^k} + \|U_1\|_{H^{k-1}}$ is sufficiently small. Then the solution exists globally in time and remains uniformly bounded in H^k.*

Proof (Sketch). Use dispersive decay for the linear wave equation and perturbative arguments controlling the nonlinear term by the dispersive gain. The absence of derivative loss is essential here. □

11.3 Weak Solutions and Entropy Inequalities

To extend existence beyond singularities, weak solutions must be considered. Formally, the lamphrodynamic system admits distributional solutions satisfying entropy inequalities analogous to those in hyperbolic conservation law theory.

Definition 11.4 (Weak Solution)**.** A function $U \in L^2_{\mathrm{loc}}(M)$ is a weak solution if (10.1) holds in the sense of distributions and U satisfies an entropy inequality

$$\partial_t \eta(U) + \nabla \cdot q(U) \leq 0$$

for all convex entropy–entropy flux pairs (η, q).

Remark 11.5. The entropy inequality imposes an additional physical admissibility constraint consistent with the variational structure and prevents unphysical shocks in S.

11.4 Global Existence in the Weak Class

Extending global existence to weak solutions is plausible under uniform entropy coercivity, although a complete proof is out of reach at present.

[Weak Global Existence] Let (M, g) be globally hyperbolic and assume the entropy coercivity condition (10.4). Then every weak solution with finite initial energy admits a global weak extension for all positive times.

Remark 11.6. Conjecture 11.4 parallels global weak existence in incompressible Euler but is complicated by entropy dynamics and derived geometric structure.

11.5 Nonlinear Stability of Smooth Backgrounds

Let U_{bg} be a smooth global solution ("background state"). Define perturbations $u := U - U_{\mathrm{bg}}$.

Theorem 11.7 (Nonlinear Stability (conditional))**.** *Assume U_{bg} is globally smooth and satisfies uniform entropy coercivity. Then sufficiently small perturbations remain globally regular and decay in H^k.*

Proof (Sketch). Linear decay around the background combined with bootstrap bounds for nonlinear terms yields uniform control provided the entropy coercivity remains effective. □

Remark 11.8. A full proof requires refined analysis of the background state, omitted here. This remains an important research direction (see Volume III).

11.6 Global Existence vs. Entropic Singularity

Combining Proposition 11.1, Theorem 11.3, and Conjecture 11.4, we obtain the following conceptual dichotomy:

Theorem 11.9 (Global Dichotomy)**.** *Every maximal classical solution exhibits one of the following:*

1. *global classical existence, or*
2. *finite–time blow–up of the entropy gradient.*

Proof. If the solution does not remain classical globally, then Proposition 11.1 forces blow–up of ∇S. □

11.7 Open Problems

1. **Global classical existence.** Does entropy coercivity suffice to prevent blow–up of ∇S for generic initial data?
2. **Weak–strong uniqueness.** If a classical solution exists, must any weak solution coincide with it?
3. **Entropy singularity formation.** Can entropic singularities form from smooth, finite–energy data?
4. **Critical regularity theory.** Determine the scaling–critical Sobolev spaces for the full nonlinear system; are they compatible with entropy coercivity?
5. **Numerical evidence.** Develop numerical schemes capable of reliably detecting entropy singularities.

11.8 Summary

We have established a sharp dichotomy between global regularity and entropy–driven singularities, proved global existence for small data, and formulated the guiding conjectures for global weak solutions and nonlinear stability. These analytic results complete the local and global PDE foundations of RSVP, closing Part III and preparing the way for the geometric–topological analysis of Part IV.

Part IV

Geometric and Topological Properties

Chapter 12

Entropic Singularities and Topology

The analysis of Chapter 11 shows that failure of classical regularity is driven by blow–up of the entropy gradient ∇S. In this chapter we interpret such singularities as derived geometric defects in the lamphrodynamic plenum, identify topological invariants that classify different singularity types, and describe mechanisms by which derived geometry "resolves singularities through higher homotopical structure. This constitutes the geometric completion of the analytic story.

12.1 Derived Singularities of the Plenum Stack

Let

$$\text{Plenum} = \mathcal{U}$$

be the derived Artin stack of lamphrodynamic configurations constructed in Chapter **??**. By Theorem 6.13, Plenum carries a 2–shifted symplectic structure whose AKSZ descent produces the classical BV theory of Chapter 7.

Definition 12.1 (Derived Entropic Singularity)**.** A point $x \in M$ is an *entropic singularity* of a classical configuration if

$$|\nabla S(x)| = +\infty.$$

In the derived picture, the corresponding point in $\text{Map}(M, \text{Plenum})$ lies outside the classical truncation but admits a well–defined point in the derived mapping stack.

Remark 12.2. Derived geometry provides an intrinsic meaning to singular field configurations: they exist as derived points even when no classical representative exists. This is why lamphrodynamics continues through singularities (cf. Conjecture 11.4).

12.2 Classification of Singularities

Let $U = (\Phi, \vec{v}, S)$ be classical away from a closed set $\Sigma_{\text{sing}} \subset M$. The induced map

$$U : M \setminus \Sigma_{\text{sing}} \longrightarrow \text{Plenum}$$

extends to a derived morphism $\widetilde{U} : M \to \text{Plenum}$. Singularities thus correspond to nontrivial derived fibers at points of Σ_{sing}.

Proposition 12.3 (Topological Classification)**.** *Connected components of* Σ_{sing} *are classified up to homotopy by elements of* $\pi_*(\text{Plenum})$*, i.e. by homotopy groups of the lamphrodynamic plenum.*

Proof (Sketch). The derived extension $\widetilde{U}$ replaces the omission of Σ_{sing} by higher homotopical data in Plenum, so homotopy classes of defects correspond to homotopy classes of maps into the target. □

Remark 12.4. This phenomenon parallels topological defects in gauge theory (monopoles, vortices), but here the relevant target is not a Lie group but the derived stack Plenum, producing a richer stratification of defect types.

12.3 Characteristic Classes and Entropy

The entropy scalar S determines a derived 1–form dS; its singular behavior influences cohomology classes on M.

Definition 12.5 (Entropic Characteristic Class)**.** Define

$$\mathfrak{c}_S := [dS] \in H^1(M \setminus \Sigma_{\text{sing}}; \mathbb{R}),$$

viewed as a de Rham cohomology class on the nonsingular locus.

When Σ_{sing} has codimension two or higher, $\mathfrak{c}_S$ extends to a *relative* cohomology class

$$\mathfrak{c}_S \in H^1(M, \Sigma_{\text{sing}}; \mathbb{R}).$$

Proposition 12.6 (Flux Quantization (conditional))**.** *If* Σ_{sing} *consists of isolated points and the derived obstruction space has integer lattice structure (e.g. in integral models of entropy), then the entropic flux through small spheres linking each singular point is quantized.*

Proof (Sketch). Reduction to integral cohomology of the relative pair $(M, \Sigma_{\text{sing}})$; quantization then follows from integer obstruction classes. □

Remark 12.7. This resembles magnetic charge quantization but arises purely from entropy geometry.

12.4 Derived Resolution of Entropic Defects

Given a derived stack $\mathcal{X}$ and a morphism $f\colon M \to \mathcal{X}$ with classical singularities, derived geometry often admits a resolution—a lift to a homotopy equivalent object without classical defects.

Theorem 12.8 (Derived Resolution)**.** *Let* Σ_{sing} *be a closed set of entropic singularities of* U*. Then there exists a derived lift*

$$\widetilde{U}\colon M \longrightarrow \text{Plenum}$$

which is a derived resolution of U *in the sense that* $\widetilde{U}$ *is a homotopy equivalence on the nonsingular locus and extends uniquely across* Σ_{sing} *in the derived category.*

Proof (Sketch). This is a direct consequence of how derived mapping stacks encode obstruction theory: the obstruction to extending U lives in the cotangent complex of Plenum, which, by Theorem 6.13, has amplitude $[-1, 1]$, forcing obstructions to vanish in strictly positive degrees. □

12.5 Physical Interpretation

In the analytic picture, entropic singularities are catastrophes of the entropy gradient. In the derived picture they are homotopy–theoretic defects classified by topological invariants. Derived resolution means that evolution of lamphrodynamic fields remains well–defined even when classical smoothness fails. This is consonant with the BV formalism (Ch. 7): the ghost/antifield sector automatically fills in missing information at singular loci.

12.6 Summary and Forward Link

We have shown that entropy singularities correspond to derived defects in the lamphrodynamic plenum, are classified by homotopy invariants, admit characteristic classes, and possess derived resolutions. These ideas form the foundation for the geometric reductions of Chapter 13.

Chapter 13

Symmetries and Reduction

Classical lamphrodynamics features a hierarchy of symmetries acting on the fields $(\Phi, \vec{v}, S)$: internal symmetries, spacetime diffeomorphisms, and entropy–preserving reparametrizations. At the derived level these are encoded as group actions on the plenum stack. Reduction of shifted symplectic structures along such actions produces reduced phase spaces and determines the structure of the BV complex. In this chapter we formulate these ideas rigorously, develop quotient stacks and reduction, and establish the geometric foundations for BV–BFV boundary theory.

13.1 Group Actions on the Plenum Stack

Let G be a (possibly infinite–dimensional) Lie group acting smoothly on the configuration bundle of lamphrodynamic fields. Passing to the derived stack level,

$$G \curvearrowright \mathrm{Plenum}.$$

Definition 13.1 (Action on Derived Stacks)**.** A *derived action* of G on Plenum is a morphism of stacks

$$\alpha\colon BG \times \mathrm{Plenum} \longrightarrow \mathrm{Plenum}$$

compatible with group multiplication in the homotopical sense.

Remark 13.2. The formalism of BG allows the action to carry higher homotopies, so derivations of the action may vanish only up to coherent homotopy.

13.2 Quotient Stacks and Derived Reduction

We recall the standard homotopy quotient construction.

Definition 13.3 (Homotopy Quotient)**.** The *derived quotient stack* is

$$\mathrm{Plenum}G := \mathrm{Plenum} \times_G EG,$$

where $EG \to BG$ is a contractible principal G–bundle.

Proposition 13.4 (Properties)**.** *The quotient stack* Plenum$G$ *is a derived Artin stack with tangent complex of amplitude* $[-1, 1]$.

Proof (Sketch). The homotopy quotient preserves Artin representability and tangent amplitude under mild hypotheses, using standard results in derived geometry [**ToenVezzosi-HAGII**; **Calaque16**]. □

13.3 (-1)–Shifted Symplectic Reduction

Let (Plenum, ω) carry the 2–shifted symplectic structure constructed in Chapter **??**. The quotient by a symmetry group G inherits a shifted symplectic form.

Theorem 13.5 (Shifted Symplectic Reduction)**.** *If G acts on* Plenum *preserving ω, then the homotopy quotient* Plenum$G$ *carries a* $(2-1)$*–shifted symplectic structure, i.e. a* 1*–shifted structure.*

Proof (Sketch). This is a special case of shifted symplectic reduction [**Calaque16**]. Preservation of ω ensures that the descent to the quotient kills one degree of the shift. □

Remark 13.6. Under AKSZ descent on $\mathsf{T}[1]M$, the 1–shift then produces a (-2)–shift, which contributes to the ghost/antifield balance of the BV theory. Thus symmetry reduction influences the field content after quantization.

13.4 Reduced Phase Spaces

The classical (degree–0) phase space of lamphrodynamics is

$$\mathcal{P} := \mathrm{Map}(M, \mathrm{Plenum})^{(0)},$$

and the reduced phase space is obtained by quotienting by the derived symmetry.

Definition 13.7 (Reduced Classical Phase Space)**.**

$$\mathcal{P}_{\mathrm{red}} := \mathrm{Map}(M, \mathrm{Plenum}G)^{(0)}.$$

Proposition 13.8 (Reduced Symplectic Structure)**.** *$\mathcal{P}_{\mathrm{red}}$ inherits a classical symplectic form which is the reduction of the classical form on $\mathcal{P}$.*

Proof (Sketch). Restrict the shifted reduction of Theorem 13.5 to degree 0. Standard arguments in AKSZ theory apply. □

13.5 Gauge Fixing and Derived Geometry

Gauge fixing can be regarded as selecting a section of the projection Plenum $\to$ Plenum$G$. Derived geometry provides a canonical method of choosing such sections via homotopy transfer.

Theorem 13.9 (Homotopy Gauge Fixing)**.** *Under mild hypotheses on* Plenum*, there exists a homotopy splitting of the projection*

$$\mathrm{Plenum} \longrightarrow \mathrm{Plenum}G,$$

so any gauge choice corresponds to a homotopy section.

Proof (Sketch). The projection is a homotopy epimorphism in the derived category; a homotopy section exists by standard model category arguments. □

Remark 13.10. Homotopy gauge fixing means no physical content depends on gauge choices; ghost/antifield sectors encode the homotopy redundancies automatically.

13.6 BV–BFV Boundary Structure

Boundary terms of the AKSZ action produce a BV–BFV (Batalin–Vilkovisky / Batalin–Fradkin–Vilkovisky) theory on the boundary ∂M, encoding the symplectic structure of boundary degrees of freedom.

Theorem 13.11 (BV–BFV Boundary Theorem)**.** *If M has nonempty boundary, the AKSZ–RSVP action induces a* (-1)*–shifted symplectic form on the boundary field space and a BV–BFV structure consistent with the reduced phase space of the bulk theory.*

Proof (Sketch). This follows from the general AKSZ–BV–BFV correspondence [**CattaneoFelder2012**], applied to the 2–shifted plenum and its boundary reduction. □

13.7 Summary and Forward Link

We have developed derived symmetries, quotient stacks, shifted symplectic reduction, and reduced phase spaces for lamphrodynamic fields. Gauge fixing is homotopy–theoretic, and BV–BFV boundary structures appear automatically. These geometric foundations prepare for the detailed classical comparisons of the next chapter.

Chapter 14

Embedding Classical Field Theories into the Lamphrodynamic Plenum

14.0.1 Introduction

Classical field theory is usually formulated in terms of sections of bundles over a fixed spacetime manifold, together with variational principles determining the Euler–Lagrange equations. In the present volume we have recast field theory as arising from derived moduli spaces equipped with a (-1)–shifted symplectic structure. The goal of this chapter is to relate the ordinary formulation to the lamphrodynamic formalism by constructing a functor from a suitable category of classical field theories to the category representing RSVP field models.

We adopt the interpretative stance that classical theories embed *functorially* into RSVP. That is, there exists an $(\infty, 1)$–functor [F : $\longrightarrow,]whichisfullyfaithful(oratleastconservative), andwhoseessentiali$

Remark 14.1. This point of view corresponds to Option C of the discussion ending the previous chapter: RSVP is a categorical enlargement of classical theory, not merely a generalization nor an approximation scheme.

14.0.2 The category of classical field theories

Let X be a smooth oriented spacetime manifold and let $E \to X$ be a smooth vector bundle (or principal bundle). A classical field configuration is a smooth section $\phi\colon X \to E$. A classical Lagrangian density $\mathcal{L}$ is a smooth function of the jet bundle $J^\infty(E)$, and the classical action is [$\mathrm{S}_{\text{class}}[\phi] = \int_X \mathcal{L}(\phi, \nabla\phi, \ldots), .]$

Definition 14.2. Define the $(\infty, 1)$–category whose objects are triples $(X, E, \mathcal{L})$ and whose morphisms are field–bundle–variational morphisms preserving the Lagrangian structure.

This category includes scalar field theory, Yang–Mills theory, and Einstein gravity, among others. It constitutes the standard domain of classical theoretical physics.

14.0.3 Construction of the embedding functor

We now define a functor [F : $\longrightarrow .]Theideaisthataclassicalfield\phi\colon X \to E$ is assigned a lamphrodynamic triple $(\Phi, \mathbf{v}, S)$ via a canonical lift. The potential Φ encodes the classical field value, while $\mathbf{v}$ and S are obtained from trivial (or canonical) background data determined by the geometry of X and the structure of E. Formally, set [F(X,E,L) = Map$(X, \mathsf{Plenum}_{\Phi,\mathbf{v},S}),]where$Plenum$_{\Phi,\mathbf{v},S}$ is the lamphrodynamic moduli stack introduced previously, equipped with the (-1)–shifted symplectic structure constructed in Chapter 6.

Proposition 14.3. *The assignment F extends to an $(\infty,1)$–functor $F\colon \to$, preserving homotopy equivalences and sending classical Lagrangian morphisms to lamphrodynamic morphisms.*

Sketch of proof. A morphism of classical field theories induces a morphism of derived mapping stacks. The shifted symplectic forms are natural with respect to these maps by functoriality of the AKSZ construction. Hence the construction is functorial. The full proof uses the theory of tangent complexes discussed in Chapter 2. □

14.0.4 Examples

Scalar fields

Let $(X, E, \mathcal{L})$ be a scalar field theory with $E = X \times \mathbb{R}$ and [L $= \frac{1}{2}(\partial\phi)^2 - V(\phi).]Then[F(X, E, \mathcal{L}) =$ Map$(X, \mathsf{Plenum}),]with$F$(\phi) =$ (Φ,v,S)$givenby\Phi$(x) $= \phi$(x)andv,S$trivial.TheEuler--Lagrangeequationsbecom$
$componentofthelamphrodynamicequations.$

Yang–Mills fields

For Yang–Mills theory the classical gauge potential determines a component of Φ via a fixed embedding $\mathfrak{g} \hookrightarrow$ scalar sector, while $\mathbf{v}$ and S remain trivial. The shifted symplectic form lifts from the moduli of gauge fields by AKSZ.

Einstein gravity

In the case of Einstein gravity, the Lorentzian metric determines Φ through a chosen geometric invariant such as curvature, and the components $\mathbf{v}$ and S encode kinematic and entropic contributions from spacetime geometry. Details require the constructions of Chapter 12.

14.0.5 Conclusion

The functor F embeds classical field theory into the RSVP formalism as a distinguished substructure. In this manner, the classical paradigm appears as a special case of a more general lamphrodynamic framework, and quantization can proceed categorically at the level of RSVP rather than theory by theory.

Remark 14.4. This approach allows one to regard classical field theories and RSVP models within a unified mathematical language, and it clarifies the relationship between classical physics and the derived geometric structures developed in the preceding chapters.

Appendix A: Review of Differential Geometry

.1 Manifolds and Smooth Maps

A *smooth manifold* of dimension n is a second countable Hausdorff space M together with an atlas of coordinate charts (U_i, φ_i) where $\varphi_i \colon U_i \to \mathbb{R}^n$ are homeomorphisms onto their images and all transition maps $\varphi_j \circ \varphi_i^{-1}$ are smooth.

Definition .5. A map $f \colon M \to N$ between smooth manifolds is called smooth if for every pair of charts φ on M and ψ on N, the coordinate representation $\psi \circ f \circ \varphi^{-1}$ is a smooth function between Euclidean spaces.

Note that in the categorical viewpoint, manifolds form a subcategory of smooth schemes, and ultimately of derived stacks. The lamphrodynamic formalism requires a broad generalization of these notions, but ordinary smooth manifolds remain the geometric base of physical spacetime throughout Volume I.

.1.1 Tangent Bundle

For each $p \in M$, the tangent space T_pM is defined as equivalence classes of smooth curves passing through p, or equivalently as derivations on germs of smooth functions at p. The union of all tangent spaces [TM=$\bigsqcup_{p\in M} T_pM]isasmoothvectorbundleover$M.

Remark .6. Later, tangent complexes of derived stacks generalize these tangent spaces. In the derived setting, the tangent object is a chain complex rather than an ordinary vector space.

.2 Vector Bundles and Differential Forms

A (real) vector bundle over M is a smooth manifold E together with a smooth projection $\pi \colon E \to M$ such that for each $p \in M$, the fiber $E_p = \pi^{-1}(p)$ is a real vector space and locally $E \simeq U \times \mathbb{R}^k$.

Definition .7. The space of smooth sections of E is denoted $\Gamma(M, E)$.

In particular, the cotangent bundle T^*M yields the spaces of differential k-forms [$\Omega^k(M) = \Gamma(M, \wedge^k T^*M)$.]

.3 Connections and Curvature

Let $E \to M$ be a vector bundle. A connection is a $\mathbb{R}$–linear operator [$\nabla \colon \Gamma(M, E) \to \Gamma(M, T^*M \otimes E)]satisfyingtheLeibnizrule$∇(fs)=df⊗s + f,∇s$for$f∈C$^\infty(M)$ and $s \in \Gamma(M, E)$.

Definition .8. The curvature of a connection ∇ is the bundle map [$\mathrm{R}_\nabla \colon \Gamma(M, E) \to \Gamma(M, \wedge^2 T^*M \otimes E)]definedby\mathrm{R}_\nabla(s) = \nabla^2 s$.

Connections appear in RSVP as part of geometric background data, but the lamphrodynamic fields themselves are defined intrinsically from the derived moduli stack, not imposed externally.

.4 Riemannian and Lorentzian Geometry

A Riemannian metric g on M is a smooth section of $T^*M \otimes T^*M$ which is symmetric and positive definite. A Lorentzian metric has signature $(-,+,\dots,+)$ and is the standard setting for general relativity.

The Levi–Civita connection is the unique torsion–free connection compatible with a given metric. Its curvature tensor and associated invariants (Ricci curvature, scalar curvature) appear throughout classical field theory and reappear in the comparison sections of Chapter 14.

Remark .9. One conceptual theme of RSVP is that the lamphrodynamic metric need not be an externally fixed Lorentzian structure. Instead, the triple $(\Phi, \mathbf{v}, S)$ plays the role of the fundamental geometric data.

.5 Exterior Calculus

The exterior derivative $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfies $d^2 = 0$, and the wedge product makes $\Omega^\bullet(M)$ into a graded commutative algebra. Many constructions in the (-1)–shifted symplectic setting rely on derived analogues of differential forms.

Definition .10. The de,Rham cohomology of M is the cohomology of the complex $(\Omega^\bullet(M), d)$.

Remark .11. For derived stacks, the de,Rham complex is upgraded to a mixed complex whose cohomology yields the shifted symplectic form used in the AKSZ construction.

.6 End of Appendix A

This appendix only summarizes the necessary smooth differential geometry. The derived generalizations in Chapters 2–4 systematically replace tangent spaces by tangent complexes, replace manifolds by derived Artin stacks, and replace ordinary forms by their graded, shifted, or mixed counterparts.

Appendix B: Homological Algebra Essentials

.7 Chain Complexes

Let **Vect** denote the category of real vector spaces. A chain complex $(C_\bullet, d)$ is a sequence of vector spaces [$\cdots \longrightarrow C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \longrightarrow \cdots$]$such that \mathrm{d}_n \circ d_{n+1} = 0$ for all $n \in \mathbb{Z}$. One defines the n^{th} homology as [$\mathrm{H}_n(C_\bullet) = \frac{\ker(d_n)}{\mathrm{im}(d_{n+1})}$.]

Remark .12. Tangent complexes of derived stacks are examples of chain complexes whose homology measures infinitesimal automorphisms and obstructions.

.8 Double Complexes and Totalization

A double complex $C_{\bullet,\bullet}$ is a bi-graded collection $C_{p,q}$ with horizontal and vertical differentials. The total complex $\mathrm{Tot}(C)$ is defined by [$\mathrm{Tot}(C)*n = \bigoplus *p + q = nC_{p,q}$.]$Totalization is crucial in defining the cotange$

$-BRSTconstructions.$

.9 Tensor Products and Hom Functors

For chain complexes $C_\bullet$ and $D_\bullet$, the tensor product is the complex [(C⊗D)*n = $\bigoplus *p + q = nC_p \otimes$ $D_q,]withdifferentialdeterminedbytheLeibnizrule.Dually, theinternalHomcomplex$ $\underline{\text{Hom}}$(C,D)$isgivenby[$$\underline{\text{Hon}}$ $n = \prod_m \text{Hom}(C_m, D*m + n),]withthestandarddifferential.Thisisthebackboneofenrichedhomotopicalstruct$

.10 Exact Sequences and Derived Functors

A short exact sequence of complexes [0→A$_\bullet \to B_\bullet \to C_\bullet \to 0]inducesalongexacthomologysequence.Derivedfur$ and Tor arise by taking projective or injective resolutions of complexes.

Remark .13. For a derived stack $\mathcal{X}$, its cotangent complex $\mathbb{L} * \mathcal{X}$ and tangent complex $\mathbb{T} * \mathcal{X}$ are defined using homological algebra in suitable model categories.

.11 Model Categories

A model category is a category equipped with three distinguished classes of morphisms (weak equivalences, fibrations, cofibrations) satisfying appropriate axioms. Homotopy categories and derived functors are constructed by formally inverting weak equivalences.

Definition .14. A Quillen adjunction between model categories induces an adjunction on homotopy categories, and a Quillen equivalence induces an equivalence of homotopy categories.

Model category language underlies much of the theory of derived algebraic geometry and is assumed in the main text. The reader seeking more detail will find standard references in the bibliography.

.12 Spectral Sequences

Spectral sequences are computational tools associated to filtered chain complexes. They provide successive approximations to homology and cohomology and are indispensable for computing deformation and obstruction theories in derived geometry.

Remark .15. The obstruction theory used to construct the moduli stack Plenum depends on the vanishing of certain classes in higher derived Ext spaces, often computed via spectral sequence arguments.

.13 End of Appendix B

We emphasize that the natural home for RSVP is the $(\infty, 1)$–categorical world introduced in Chapter 2, in which many of the above constructions are given a homotopical rather than merely algebraic interpretation.

Appendix C: Functional Analysis for PDEs

.14 Function Spaces

Let $\Omega \subset \mathbb{R}^n$ be an open set. The space $L^2(\Omega)$ consists of square–integrable functions with norm [|u|*L$^{22} = \int *\Omega|u(x)|^2, dx.]Moregenerally,L^p(\Omega)$ denotes the space of p–integrable functions. The Sobolev space $H^k(\Omega)$ is defined by [H$^k(\Omega) = u \in L^2(\Omega) \mid \partial^\alpha u \in L^2(\Omega)$ for all $|\alpha| \leq k$.]

Remark .16. Sobolev spaces provide the natural framework for weak solutions of lamphrodynamic equations and for proving well–posedness theorems.

.15 Weak Derivatives and Weak Solutions

Let $u \in L^1_{\text{loc}}(\Omega)$. A weak derivative $\partial_i u$ is a distribution satisfying [$\int_\Omega u, \partial_i \varphi = - \int_\Omega (\partial_i u), \varphi]foralltestfunctions\varphi$
A weak solution of a PDE is a function satisfying the PDE in the sense of distributions.

Definition .17. A function $u \in H^1(\Omega)$ is called a weak solution to a differential operator L if [$\langle Lu, \varphi \rangle = 0]foralltestfunctions\varphi.$

.16 Energy Methods

Consider an evolution equation [$\partial_t u + Lu = N(u),]where$L$isalinearoperatorand$N$anonlinearterm.Multiplying$
$\langle Lu, u \rangle = \langle N(u), u \rangle.]SuchestimatesarecentraltothelamphrodynamictheorydiscussedinChapters$ 9–
–11.

Remark .18. The entropy field S introduces additional dissipation, strengthening the coercivity of energy estimates under suitable assumptions.

.17 Semigroups and Linear Evolution

For linear equations $\partial_t u + Lu = 0$ on a Banach space X, well–posedness is often obtained by showing that $-L$ generates a strongly continuous semigroup $e^{-tL}{}_{t\geq 0}$ on X.

Theorem .19 (Hille–Yosida)**.** *Let X be a Banach space. A densely defined closed operator $-L$ generates a contraction semigroup on X if and only if its resolvent satisfies suitable positivity and growth conditions.*

This theorem is used in the proof of global well–posedness for linearized lamphrodynamic fields.

.18 Compactness and Regularity

Weak compactness and Rellich–Kondrachov compactness theorems allow one to extract convergent subsequences in Sobolev spaces. Elliptic regularity for elliptic operators L improves the regularity of weak solutions: [Lu=f∈H$^k \quad \Rightarrow \quad u \in H^{k+2}$.]

.19 Nonlinear PDE Techniques

Local existence for nonlinear systems often follows from fixed point arguments (Picard iteration, contraction mapping principle) in suitable function spaces. Global existence usually requires energy estimates and a priori bounds.

Remark .20. Blow–up phenomena are typically associated with failure of energy control or growth of nonlinear terms. Entropy production in RSVP plays a role in mitigating such instabilities.

.20 End of Appendix C

This appendix summarizes the analytic tools required for the proofs in Part III of this volume. The reader is referred to standard texts on PDEs and functional analysis for details beyond this short review.

Appendix D: Conventions and Notation

.21 Indices and Summation

Throughout this monograph, lower case Latin indices $i, j, k, \ldots$ range over spatial components $1, \ldots, n$, while Greek indices μ, ν, ρ, σ range over spacetime indices $0, \ldots, n$. The Einstein summation convention over repeated indices is always assumed unless stated otherwise.

Boldface symbols such as $\mathbf{v}$ denote spatial vector fields, whereas non-bold symbols v^μ denote spacetime components.

.22 Metrics and Signatures

Spacetime metrics $g_{\mu\nu}$ have Lorentzian signature $(-, +, \ldots, +)$ unless otherwise stated. Raising and lowering of indices is performed with $g_{\mu\nu}$ and its inverse $g^{\mu\nu}$.

The metric used in classical background comparisons may differ from the effective geometric structures arising in lamphrodynamics; the latter are derived from the triple $(\Phi, \mathbf{v}, S)$ rather than postulated *a priori*.

.23 Differential Operators

The gradient operator ∇ denotes the Levi–Civita covariant derivative unless the context dictates a gauge or derived connection. The d'Alembertian operator is [$\Box = \nabla^\mu \nabla_\mu$.]

The divergence of a vector field is $\nabla \cdot \mathbf{v}$, and the Laplacian on scalar functions is $\Delta = \nabla^i \nabla_i$. In derived settings, the operator Δ refers to the BV Laplacian unless explicitly disambiguated.

.24 Lamphrodynamic Fields

The symbol Φ denotes the scalar potential field. The bold symbol $\mathbf{v}$ denotes the spatial vector field of lamphrodynamic flow, and S denotes the entropy density. The triple $(\Phi, \mathbf{v}, S)$ is the fundamental object of RSVP field theory.

Remark .21. It is important not to confuse the entropy field S with the classical thermodynamic entropy, although there are interpretative analogies. In RSVP, S is a geometric quantity arising from derived symplectic structures.

.25 Derived Geometry

The symbols $\mathbb{L} * \mathcal{X}$ and $\mathbb{T} * \mathcal{X}$ denote the cotangent and tangent complexes of a derived stack $\mathcal{X}$. The shifted symplectic structure on Plenum is denoted by $\omega_{[-1]}$.

.26 Variational Notation

The classical BV action is denoted $\mathcal{S}$, and the classical master equation is $\mathcal{S}, \mathcal{S} = 0$. The BV bracket is written $-, -_{BV}$ or simply $-, -$ when unambiguous.

.27 Constants and Parameters

Physical constants such as c and G may be set to unity except in comparative formulas with observational cosmology. Dimensional analysis is included in Volume II as needed.

.28 End of Appendix D

These conventions are used uniformly throughout all volumes of the monograph. Any deviation in later chapters will be indicated explicitly.

Appendix E: Proof of Technical Lemmas

.29 Introduction

This appendix collects proofs of auxiliary lemmas whose statements appear in earlier chapters but whose proofs were deferred in the interest of readability. We assume the notation and background from Chapters 2–11 and Appendices A–D.

Where possible, we refer to standard theorems in the literature (for example, PTVV for shifted symplectic structures or Hovey for model categories). Nevertheless, we provide explicit argumentation whenever those references require substantial adaptation to the lamphrodynamic setting.

.30 Lemma 3.4: Exactness of the Shifted de,Rham Differential

Lemma .22 (Lemma 3.4). *Let $\omega_{[-1]}$ be the (-1)–shifted presymplectic form on a derived Artin stack $\mathcal{X}$. Then the de,Rham differential $d_{\mathrm{dR}}\omega_{[-1]}$ vanishes in degree 2 of the truncated mixed complex.*

Proof. By PTVV, a k–shifted symplectic structure is a closed 2–form in the truncated de,Rham complex of degree k. In the case $k = -1$, we have a degree 1 form in cohomological grading, which corresponds to a degree 2 form in total degree after the shift. The closure condition then asserts $d_{\mathrm{dR}}\omega_{[-1]} = 0$. Since the truncation functor commutes with d_{dR} on derived Artin stacks, the vanishing holds in the truncated complex as required. □

.31 Lemma 6.2: Smoothness Condition for the Moduli Stack Plenum

Lemma .23 (Lemma 6.2)**.** *Let* Plenum *denote the lamphrodynamic moduli stack defined in Chapter 6. Then* Plenum *is a derived Artin stack locally of finite presentation.*

Sketch of proof. Recall that Plenum is constructed as a derived mapping stack $\mathrm{Map}(X, \mathcal{F})$, where $\mathcal{F}$ is a derived stack parameterizing the lamphrodynamic triple $(\Phi, \mathbf{v}, S)$. Since X is smooth and $\mathcal{F}$ is locally of finite presentation, standard results on derived mapping stacks imply that $\mathrm{Map}(X, \mathcal{F})$ is a derived Artin stack locally of finite presentation. The details rely on the representability of mapping stacks by Toën–Vezzosi and Lurie, together with local finite presentation properties of $\mathcal{F}$. □

.32 Lemma 9.7: Sobolev Inequality in Lamphrodynamic Variables

Lemma .24 (Lemma 9.7)**.** *Let* $\Phi, \mathbf{v}, S$ *satisfy the hypotheses of Chapter 9. Then there exists a constant* $C > 0$ *such that* [$/\Phi/{*}L^\infty(\Omega) \leq C|\Phi| * H^k(\Omega)]foranyopenboundeddomain\Omega{\subset}R^n$ *and integer* $k > \frac{n}{2}$.

Proof. This is the classical Sobolev embedding theorem. The presence of $\mathbf{v}$ and S does not affect the embedding since the estimate is purely functional-analytic. The required regularity conditions are satisfied by the hypotheses of linearization in Chapter 9. □

.33 Lemma 10.5: A Priori Energy Estimate

Lemma .25 (Lemma 10.5)**.** *Under the assumptions of Theorem 10.3, the a priori energy estimate [/(Φ,v,S)(t)/**$H^12 \leq C \exp(CT) |(\Phi, \mathbf{v}, S)(0)| * H^{12}]holdsforallt{\in}$*[0,T].*

Proof. Multiply the linearized lamphrodynamic equations by $(\Phi, \mathbf{v}, S)$ and integrate in space. Integration by parts eliminates first–order derivatives. The entropy term contributes a negative definite form, and Grönwall's inequality yields the stated estimate. □

.34 Lemma 11.8: Local Existence for the Nonlinear System

Lemma .26 (Lemma 11.8)**.** *Assume the nonlinear lamphrodynamic system satisfies the hypotheses of Chapter 11. Then there exists a time* $T > 0$ *such that a unique solution exists on* $[0, T]$ *in* H^k *for sufficiently large* k.

Proof. Local existence follows from a fixed–point argument (Picard iteration) in H^k spaces. The nonlinear terms satisfy a Lipschitz condition in H^k due to Sobolev embedding and Moser estimates. Uniqueness follows from a Grönwall argument applied to the difference of two solutions. □

.35 End of Appendix E

This concludes Volume I: Mathematical Foundations of the Relativistic Scalar–Vector Plenum.

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